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POP-UP RETENTION MECHANISM FOR SECURELY CHARGING SMART GLASSES IN A SELF-ASSEMBLING CHARGING CASE, AND SYSTEMS OF USE THEREOF

Abstract

An example collapsible charging case for a pair of smart glasses comprises a foldable component that is configured to operate in two states, including: a folded state that defines a first internal volume, and an unfolded state that defines a second internal volume that is greater than the first interior volume, and the second interior volume being configured to house a pair of smart glasses. The collapsible charging case also includes a deployable mechanism configured to contact a nose bridge portion of a pair of smart glasses, and the deployable mechanism is configured to operate in two states, including: an undeployed state that occurs when the foldable component of the collapsible charging case is in the first foldable state, and a deployed state that occurs when the foldable component of the collapsible charging case is in the second foldable state.

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Background/Summary

RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application Ser. No. 63/561,606 filed Mar. 5, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This relates generally to collapsible charging case that is configured to house and charge a pair of smart glasses (e.g., a pair of augmented-reality glasses or a pair of smart glasses), and the collapsible charging case is configured such that it has an overall volume when the smart glasses are stored within it, and another overall volume that is less than the overall volume when the smart glasses are not stored within it.

BACKGROUND

[0003] Portable electronic devices that do not typically include large batteries (e.g., headphones) typically benefit from carrying cases that include additional batteries that can be used to charge the portable electronic device on the go (e.g., away from a power source). However, for larger portable electronic devices these charging cases can prove to be too bulky and cumbersome to carry around thereby reducing the chances that an end user brings the charging case with them on the go. As such, bulky and cumbersome charging cases can result in user's not making intermittent charges to the portable electronic device thereby reducing the overall use time of the device while on the go.

[0004] As such, there is a need to address one or more of the above-identified challenges. A brief summary of solutions to the issues noted above are described below.

SUMMARY

[0005] The collapsible charging case described herein solves the issues noted above, notably by making the collapsible charging case capable of reducing its overall volume when not storing and charging a pair of smart glasses. The benefit of the collapsible charging case changing its volume when not being used to charge or store a pair of smart glasses allows for a more convenient form factor for carrying on the go (e.g., able to fit in a pocket or not take up volume in a bag).

[0006] In one example collapsible charging case for a pair of smart glasses (e.g., a collapsible charging case shown in FIGS. 1A-3) described herein comprises a foldable component of the collapsible charging case. The collapsible charging case is configured to operate in at least two states (e.g., the foldable component is hingeably coupled to the charging case, and can include a spring (e.g., a torsion spring) to automatically transition between the at least two states) (e.g., four rectangular pieces **116A-116D** and two triangular pieces **118A** and **118B** shown in FIG. 1). The at least two states include (i) a folded state that partially defines a first internal volume (e.g., a collapsed state as described in reference to FIG. 1), and (ii) an unfolded state that partially defines a second internal volume that is greater than the first interior volume, and the second interior volume being configured to house a pair of smart glasses (e.g., a non-collapsed state as described in reference to FIG. 1) (e.g., the second state is configured such that the foldable component is positioned to form a right angle with (e.g., perpendicular to) another surface of the collapsible charging case). The collapsible charging case also comprises a deployable mechanism configured to contact a nose bridge portion of a pair of smart glasses (e.g., deployable mechanism **114** shown

in FIG. 1 and deployable mechanism 202 shown in FIG. 2) (e.g., the nose bridge portion of the pair of smart glasses includes one or more electrical contacts that are configured to receive charge from another set of one or more electrical contacts located on the deployable mechanism), and the deployable mechanism is configured to operate in at least two states (e.g., FIG. 2 shows the deployable mechanism operating in two different states, a deployed state and a non-deployed state). These two states are (i) an undeployed state that occurs when the foldable component of the collapsible charging case is in the first foldable state (FIG. 2 shows in a second pane 238 that the deployable mechanism 202 is in its fully non-deployed state), and a deployed state that occurs when the foldable component of the collapsible charging case is in the second foldable state, wherein the deployable mechanism is configured to contact a nose bridge portion of the pair of smart glasses in the deployed state (e.g., FIG. 2 shows in a first pane 200 that the deployable mechanism 202 is in its fully deployed state).

[0007] The features and advantages described in the specification are not necessarily all inclusive and, in particular, certain additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes.

[0008] Having summarized the above example aspects, a brief description of the drawings will now be presented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] For a better understanding of the various described embodiments, reference should be made to the Detailed Description below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

[0010] FIGS. 1A and 1B illustrate a collapsible charging case for a pair of smart glasses that includes a deployable mechanism that is configured to contact a nose bridge portion of a part of smart glasses, in accordance with some embodiments.

[0011] FIG. 2 illustrates a deployable mechanism transitioning from the deployed state to the non-deployed state when the collapsible charging case transitions from the non-collapsed state to the collapsed state, and vice versa, in accordance with some embodiments.

[0012] FIG. 3 illustrates a portion of the collapsible charging case that includes the deployable mechanism, an on-board device (e.g., a Lithium-ion battery), and one or more electrical components for facilitating charging a pair of smart glasses and the on-board battery, in accordance with some embodiments.

[0013] FIGS. 4A, 4B, 4C-1, 4C-2, 4D-1, and 4D-2 illustrate example artificial-reality systems, in accordance with some embodiments.

[0014] FIGS. 5A-5B illustrate an example wrist-wearable device 500, in accordance with some embodiments.

[0015] FIGS. 6A, 6B-1, 6B-2, and 6C illustrate example head-wearable devices, in accordance with some embodiments.

[0016] FIGS. 7A-7B illustrate an example handheld intermediary processing device, in accordance with some embodiments.

[0017] FIGS. 8A-8C illustrate an example smart textile-based garment, in accordance with some embodiments.

[0018] FIG. 9 illustrates a multi-dimensional knitting machine configured to produce multi-dimensional knitted smart textile-based garments in an automated fashion, in accordance with some embodiments.

[0019] FIG. **10** illustrates an alternative deployable mechanism used for securing a pair of smart glasses to the collapsible charging case, in accordance with some embodiments.

[0020] FIG. **11** shows a progression of the deployable mechanism transitioning between a closed state, an open state, and an open state with a pair of smart glasses inserted, in accordance with some embodiments.

[0021] FIGS. **12A-12C** illustrate three types of deployable mechanisms for controlling the movement of the portion that retains a pair of smart glasses, in accordance with some embodiments.

[0022] FIG. **13** illustrates how the collapsible charging case that uses strategically placed magnets to cause the collapsible charging case to “automatically” open once it is partially opened by a user, in accordance with some embodiments.

[0023] FIG. **14** illustrates cutaway views and of a wing rotating about a bottom portion **1404**, in accordance with some embodiments.

[0024] In accordance with common practice, the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method, or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

DETAILED DESCRIPTION

[0025] Numerous details are described herein to provide a thorough understanding of the example embodiments illustrated in the accompanying drawings. However, some embodiments may be practiced without many of the specific details, and the scope of the claims is only limited by those features and aspects specifically recited in the claims. Furthermore, well-known processes, components, and materials have not necessarily been described in exhaustive detail so as to avoid obscuring pertinent aspects of the embodiments described herein.

[0026] Embodiments of this disclosure can include or be implemented in conjunction with various types or embodiments of artificial-reality systems. Artificial-reality (AR), as described herein, is any superimposed functionality and or sensory-detectable presentation provided by an artificial-reality system within a user's physical surroundings. Such artificial-realities can include and/or represent virtual reality (VR), augmented reality, mixed artificial-reality (MAR), or some combination and/or variation one of these. For example, a user can perform a swiping in-air hand gesture to cause a song to be skipped by a song-providing API providing playback at, for example, a home speaker. An AR environment, as described herein, includes, but is not limited to, VR environments (including non-immersive, semi-immersive, and fully immersive VR environments); augmented-reality environments (including marker-based augmented-reality environments, markerless augmented-reality environments, location-based augmented-reality environments, and projection-based augmented-reality environments); hybrid reality; and other types of mixed-reality environments.

[0027] Artificial-reality content can include completely generated content or generated content combined with captured (e.g., real-world) content. The artificial-reality content can include video, audio, haptic events, or some combination thereof, any of which can be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional effect to a viewer). Additionally, in some embodiments, artificial reality can also be associated with applications, products, accessories, services, or some combination thereof, which are used, for example, to create content in an artificial reality and/or are otherwise used in (e.g., to perform activities in) an artificial reality.

[0028] A hand gesture, as described herein, can include an in-air gesture, a surface-contact gesture, and or other gestures that can be detected and determined based on movements of a single hand (e.g., a one-handed gesture performed with a user's hand that is detected by one or more sensors of a wearable device (e.g., electromyography (EMG) and/or inertial measurement units (IMU) s of a

wrist-wearable device) and/or detected via image data captured by an imaging device of a wearable device (e.g., a camera of a head-wearable device)) or a combination of the user's hands. In-air means, in some embodiments, that the user hand does not contact a surface, object, or portion of an electronic device (e.g., a head-wearable device or other communicatively coupled device, such as the wrist-wearable device), in other words the gesture is performed in open air in 3D space and without contacting a surface, an object, or an electronic device. Surface-contact gestures (contacts at a surface, object, body part of the user, or electronic device) more generally are also contemplated in which a contact (or an intention to contact) is detected at a surface (e.g., a single or double finger tap on a table, on a user's hand or another finger, on the user's leg, a couch, a steering wheel, etc.). The different hand gestures disclosed herein can be detected using image data and/or sensor data (e.g., neuromuscular signals sensed by one or more biopotential sensors (e.g., EMG sensors) or other types of data from other sensors, such as proximity sensors, time-of-flight (ToF) sensors, sensors of an inertial measurement unit, etc.) detected by a wearable device worn by the user and/or other electronic devices in the user's possession (e.g., smartphones, laptops, imaging devices, intermediary devices, and/or other devices described herein).

[0029] A collapsible charging case configured to charge a pair of glasses (e.g., smart glasses, extended reality glasses (e.g., AR, VR, MR glasses) are described herein. The collapsible charging case is designed to have an appearance of a traditional glasses case and provide an end-user with a familiar user experience (e.g., placing glasses in a case when not in use).

[0030] FIGS. 1A and 1B illustrate a collapsible charging case for a pair of smart glasses that includes a deployable mechanism that is configured to contact a nose bridge portion of a part of smart glasses, in accordance with some embodiments. FIG. 1A shows a sequence **100** that illustrates how the collapsible charging case **102** transitions from its non-collapsed state that is configured to house a pair of smart glasses (occluded) within an interior space and charge a battery included with the pair of smart glasses to a collapsed state. In the first sequence **104**, the collapsible charging case **102** is shown in the non-collapsed state **106** that is configured to house and charge a pair of smart glasses. In some embodiments, the non-collapsed state is maintained through the use of magnets to allow the collapsible charging case to maintain its shape. In some embodiments, the non-collapsed state is maintained through the use of hook and loop fasteners, either alone or in conjunction with magnets.

[0031] The second sequence **108** shows an intermediary state **110** in which the collapsible charging case **102** is transitioning from the non-collapsed state **106** to the collapsed state **112**. As shown in the intermediary state **110** the collapsible charging case **102** is shown to include a deployable mechanism **114** configured to contact a nose bridge portion of a pair of smart glasses (discussed in further detail in reference to FIG. 2). In some embodiments, the deployable mechanism **114** is spring loaded such that it can be in a position to better contact the pair of smart glasses when the collapsible charging case **102** is in the non-collapsed state and another position that is flattened when the collapsible charging case **102** is in the collapsed state to improve portability when being transported (i.e., a smaller volume in the collapsed state improves comfort and convenience when transporting in pockets or bags). The second sequence **108** also shows that there are at least six movable portions relative to each other, and those moveable portions include four rectangular pieces **116A-116D** and two triangular pieces **118A** and **118B**. In some embodiments, the two triangular pieces **118A** and **118B** are hinged and automatically flip open to provide structure.

[0032] FIG. 1A also shows a third sequence **120** where the collapsible charging case **102** in the collapsed state **112** where the pair of smart glasses are not in the collapsible charging case **102** and the deployable mechanism **114** (shown in FIG. 2) is in its undeployed state which reduces the volume in which the deployable mechanism **114** occupies. In some embodiments, collapsible charging case **102** is maintained in the non-deployed state through the use of magnets. In some embodiments, the magnets used to keep the collapsible charging case **102** in its collapsed state are the same magnets that are used to keep the collapsible charging case **102** in the non-collapsed state.

[0033] FIG. 1B illustrates a sequence of how the collapsible charging case **102** enters the non-collapsed state **124** from the collapsed state **126**. In some embodiments, the two triangular pieces **118A** and **118B** are hinged and automatically flip open to provide structure when collapsible charging case **102** is opened up by the user. In some embodiments, the magnets keep the two triangular pieces in the non-deployed state and once the magnets are separated (e.g., by the user starting to open the collapsible charging case **102**) do the springs release their compressed energy to cause the collapsible charging case **102** to automatically deploy to the non-collapsed state. The magnets are also placed in certain locations, such that at least one of the four rectangular pieces **116A-116D** move to connect to deployed triangular pieces (e.g., **116D** moves to connect to the deployed triangular pieces **118A-118B**).

[0034] FIG. 2 illustrates a deployable mechanism transitioning from the deployed state to the non-deployed state when the collapsible charging case transitions from the non-collapsed state to the collapsed state, and vice versa, in accordance with some embodiments. FIG. 2 shows in a first pane **200** that the deployable mechanism **202** is in its fully deployed state and protrudes from a surface **204** by a predefined distance **205** (e.g., 5-15 mm) such that the charging contacts **206A** and **206B** can contact corresponding charging contacts located on a nose bridge of a pair of smart glasses. As illustrated by the first pane **200**, the deployed state occurs when the collapsible charging case **208** is in either a non-collapsed state **210** or in an intermediary state **212**.

[0035] As shown in the first pane, a cutaway view **214** is shown, which shows an example system the controls the movement of the deployable mechanism. As shown the deployable mechanism comprises a hinged member **216** that includes a portion **218** at the distal end of the hinged member configured to secure a pair of smart glasses within the collapsible charging case **208** (e.g., by using spring loaded contacts **220A** and **220B** (occluded) that slide over a pair of nose pads on a pair of smart glasses). This portion **218** is also configured to contact a charging contact sub assembly **222** and is configured to slide about the charging sub assembly **222** to allow for movement between the portion **218** and the charging sub-assembly **222**.

[0036] The sub-assembly **222** includes a charging contact portion **224** that includes charging contacts **220A** and **220B** configured to provide a charge to a pair of smart glasses. The sub-assembly includes a two rotatable arms **226A** and **226B** that rotatably connect the charging contact portion **224** to a fixed component **228** of the collapsible charging case **208**. The sub-assembly also includes one or more torsion spring(s) **230** that bias the deployable mechanism **202** into its deployed state. While a torsion spring is shown, other types of springs would be interchangeable, such as coil springs, leaf springs, etc. In some embodiments, a non-traditional biasing system can be conceived, e.g., through the use of materials with certain elastic characteristics, motorized systems, pressurized struts, etc.

[0037] The deployable mechanism **202** also includes a rotation limiter **234** that is configured to limit the distance in which the hinged member can extend from the surface **236**.

[0038] The first pane **200** also shows that a portion of deployable mechanism is surrounded by a flexible cover **232** (e.g., silicone, fabric, etc.) that is configured to cover the mechanisms of the deployable mechanism, which reduces dust and debris from entering the mechanisms while also providing an aesthetically pleasing appearance (i.e., covering the mechanisms from being seen by the end user).

[0039] FIG. 2 shows in a second pane **238** that the deployable mechanism **202** is in its fully non-deployed state and does not protrude from a surface **204** or slightly protrudes from surface **204** (e.g., 1-5 mm) such that collapsible charging case **208** can be collapsed and reduce its overall volume, as discussed in reference to FIG. 1. As illustrated by the second pane, the non-deployed state occurs when the collapsible charging case **208** is in a collapsed state **240**.

[0040] As shown in the second pane **238**, a cutaway view **242** is shown, which shows the example system the controls the movement of the deployable mechanism in the non-deployed state.

Cutaway view **242** illustrates that the two rotatable arms **226A** and **226B** have a different position

that is substantially parallel to fixed component **228** of the collapsible charging case **208**. In this state the torsion spring(s) **230** is in its compressed state and contains additional spring potential energy for biasing the deployable mechanism **202** into its deployed state.

[0041] The second pane **238** also shows that flexible cover **232** (e.g., silicone, fabric, etc.) elastically deforms as deployable mechanism **202** transitions to the non-deployed state. In some embodiments, the flexible cover **232** is configured to partially bias the deployable mechanism **202** into the deployed state.

[0042] FIG. **3** illustrates a portion **300** of the collapsible charging case **302** that includes the deployable mechanism **304**, an on-board device **306** (e.g., a Lithium-ion battery), and one or more electrical components **308** for facilitating charging a pair of smart glasses and the on-board battery, in accordance with some embodiments. FIG. **3** also further illustrates the portion **310** of the deployable mechanism **304** that is configured to hold a pair of smart glasses in place and in contact with the charging pins **312A** and **312B**. In some embodiments, the one or more electrical components **308** can include compute components, such that the collapsible charging case **302** acts as an intermediary processing device for the pair of smart glasses that are normally housed in the collapsible charging case **302**. See, e.g., FIGS. **7A** and **7B** for further details on an intermediary processing device. In other words, when the user has donned the pair of smart glasses the collapsible charging case **302** acts as an intermediary processing device. In some embodiments, this mode is activated in response to the detection that the smart glasses are removed from the housing and/or when the collapsible charging case **302** is in the collapsed state. In some embodiments, the collapsible charging case **302** can include one or more controls for interacting with an extended-reality displayed at the pair of smart glasses.

[0043] (A1) In accordance with some embodiments, a collapsible charging case for a pair of smart glasses (e.g., a collapsible charging case shown in FIGS. **1A-3**) comprises a foldable component of the collapsible charging case. The collapsible charging case is configured to operate in at least two states (e.g., the foldable component is hingeably coupled to the charging case, and can include a spring (e.g., a torsion spring) to automatically transition between the at least two states) (e.g., four rectangular pieces **116A-116D** and two triangular pieces **118A** and **118B** shown in FIG. **1**). The at least two states include (i) a folded state that partially defines a first internal volume (e.g., a collapsed state as described in reference to FIG. **1**), and (ii) an unfolded state that partially defines a second internal volume that is greater than the first interior volume, and the second interior volume being configured to house a pair of smart glasses (e.g., a non-collapsed state as described in reference to FIG. **1**) (e.g., the second state is configured such that the foldable component is positioned to form a right angle with (e.g., perpendicular to) another surface of the collapsible charging case). The collapsible charging case also comprises a deployable mechanism configured to contact a nose bridge portion of a pair of smart glasses (e.g., deployable mechanism **114** shown in FIG. **1** and deployable mechanism **202** shown in FIG. **2**) (e.g., the nose bridge portion of the pair of smart glasses includes one or more electrical contacts that are configured to receive charge from another set of one or more electrical contacts located on the deployable mechanism), and the deployable mechanism is configured to operate in at least two states (e.g., FIG. **2** shows the deployable mechanism operating in two different states, a deployed state and a non-deployed state). These two states are (i) an undeployed state that occurs when the foldable component of the collapsible charging case is in the first foldable state (FIG. **2** shows in a second pane **238** that the deployable mechanism **202** is in its fully non-deployed state), and a deployed state that occurs when the foldable component of the collapsible charging case is in the second foldable state, wherein the deployable mechanism is configured to contact a nose bridge portion of the pair of smart glasses in the deployed state (e.g., FIG. **2** shows in a first pane **200** that the deployable mechanism **202** is in its fully deployed state). The components described in any one of A1 through B2 are applicable to the collapsible charging case and deployable mechanisms described in reference to FIGS. **10-15**.

[0044] In some embodiments, the collapsible charging case includes one or more electrical components for acting as an intermediary processing device to reduce computational load on the pair of smart glasses. By offloading operations to the collapsible charging case, the pair of smart glasses can operate for an extended period of time as compared to a pair of smart glasses that do not offload operations.

[0045] (A2) In some embodiments of A1, the collapsible charging case for the pair of smart glasses comprises another foldable component that is on an opposite side of collapsible charging case relative to the foldable component, wherein the other foldable component mirrors a shape of the foldable component (e.g., the other foldable component is hingeably coupled to the charging case, and can include a spring (e.g., a torsion spring) to automatically transition between the at least two states). For example, FIG. 1 shows that four rectangular pieces **116A-116D** have similar or the same shapes and two triangular pieces **118A** and **118B** have similar or the same shapes.

[0046] (A3) In some embodiments of any of A1-A2, the collapsible charging case for the pair of smart glasses comprises another foldable component that is perpendicular to the foldable component, and the other foldable component is configured to cause the foldable component to enter the unfolded state when the other foldable component enters the unfolded state. In some embodiments, the other foldable component has a larger surface area than a surface area of the foldable component. For example, FIG. 1 shows that four rectangular pieces **116A-116D** and two triangular pieces **118A** and **118B**. **2**

[0047] (A4) In some embodiments of any of A1-A3, the collapsible charging case for the pair of smart glasses, wherein a shape of the other foldable component is different than the shape of the foldable component (e.g., the shape of the other foldable component is rectangular and the shape of the foldable component is triangular). For example, FIG. 1 shows that four rectangular pieces **116A-116D** and two triangular pieces **118A** and **118B**.

[0048] (A5) In some embodiments of any of A1-A4, the deployable mechanism is configured to enter the deployed state in response to the other foldable component entering the unfolded state. For example, magnets on the other foldable component can cause the deployable mechanism to be drawn into the deployed state.

[0049] (A6) In some embodiments of any of A1-A5, the other foldable component has a rectangular shape. For example, FIG. 1 shows four rectangular pieces **116A-116D**.

[0050] (A7) In some embodiments of any of A1-A6, the deployable mechanism includes a spring and the spring is in an compressed state when the foldable component is in the folded state and an uncompressed state when in the unfolded state. For example, FIG. 2 shows that the torsion spring **320** is in a uncompressed or semi-uncompressed state when the deployable mechanism **202** is in the deployed state and the torsion spring **320** is in the compressed state when the deployable mechanism **202** is in the non-deployed state.

[0051] (A8) In some embodiments of A7, the spring is a torsion spring (e.g., torsion spring **320** shown in FIG. 2).

[0052] (A9) In some embodiments of A7, the spring is a leaf spring.

[0053] (A10) In some embodiments of any of A1-A9, deployable mechanism includes a retention bracket, and the retention bracket is configured interface with nose pads of a pair of smart glasses to retain the smart glasses within the collapsible charging case. For example, FIG. 2 illustrates that a portion **218** at the distal end of the hinged member configured to secure a pair of smart glasses within the collapsible charging case **208**.

[0054] (A11) In some embodiments of any of A1-A10, the deployable mechanism includes a nose bridge interface portion that includes one or more charging contacts, and the nose bridge is configured to contact the pair of smart glasses. FIG. 2 shows spring loaded contacts **220A** and **220B** being part of the deployable mechanism, and FIG. 3 also shows charging pins **312A** and **312B**.

[0055] (A12) In some embodiments of any of A1-A11, the deployable mechanism is partially

covered by a silicone cover. For example, FIG. 2 shows in the first pane **200** that a portion of deployable mechanism is surrounded by a flexible cover **232** (e.g., silicone, fabric, etc.) that is configured to cover the mechanisms of the deployable mechanism.

[0056] (A13) In some embodiments of any of A1-A12, the collapsible charging case includes a battery and one or more electrical components configured for charging a pair of smart glasses. For example, FIG. 3 illustrates a portion **300** of the collapsible charging case **302** that includes the deployable mechanism **304**, an on-board device **306** (e.g., a Lithium-ion battery), and one or more electrical components **308**.

[0057] (A14) In some embodiments of any of A1-A13, the folded state has a cross-sectional shape that is rectangular (e.g., a rounded rectangular shape). FIG. 2 shows that the collapsible charging case **102** when in the collapsed state **106** has a rectangular or rounded rectangular shape.

[0058] (A15) In some embodiments of any of A1-A14, the unfolded state has a cross-section shape that is triangular (e.g., a rounded triangular shape). For example, FIG. 2 shows that the collapsible charging case **102** when in the non-collapsed state **106** has a triangular or rounded-triangular shape.

[0059] (A16) In some embodiments of any of A1-A15, the pair of smart glasses are a pair of augmented reality glasses. For example, FIG. 2 shows that the collapsible charging case **102** when in the collapsed state **112** has a rectangular or rounded-rectangular shape.

[0060] (A17) In some embodiments of any of A1-A16, the collapsible charging case includes one or more magnets to maintain a shape of the folded state and the unfolded state.

[0061] (B1) In accordance with some embodiments, a system includes at least (i) a collapsible charging case and (ii) a pair of smart glasses, wherein the collapsible charging case is configured in accordance with any of A1-A17.

[0062] (B2) In some embodiments of B1, the smart glasses are extended-reality glasses.

[0063] The devices described above are further detailed below, including systems, wrist-wearable devices, headset devices, and smart textile-based garments. Specific operations described above may occur as a result of specific hardware, such hardware is described in further detail below. The devices described below are not limiting and features on these devices can be removed or additional features can be added to these devices. The different devices can include one or more analogous hardware components. For brevity, analogous devices and components are described below. Any differences in the devices and components are described below in their respective sections.

[0064] As described herein, a processor (e.g., a central processing unit (CPU) or microcontroller unit (MCU)), is an electronic component that is responsible for executing instructions and controlling the operation of an electronic device (e.g., a wrist-wearable device **500**, a head-wearable device, an HIPD **700**, a smart textile-based garment **800**, or other computer system). There are various types of processors that may be used interchangeably or specifically required by embodiments described herein. For example, a processor may be (i) a general processor designed to perform a wide range of tasks, such as running software applications, managing operating systems, and performing arithmetic and logical operations; (ii) a microcontroller designed for specific tasks such as controlling electronic devices, sensors, and motors; (iii) a graphics processing unit (GPU) designed to accelerate the creation and rendering of images, videos, and animations (e.g., virtual-reality animations, such as three-dimensional modeling); (iv) a field-programmable gate array (FPGA) that can be programmed and reconfigured after manufacturing and/or customized to perform specific tasks, such as signal processing, cryptography, and machine learning; (v) a digital signal processor (DSP) designed to perform mathematical operations on signals such as audio, video, and radio waves. One of skill in the art will understand that one or more processors of one or more electronic devices may be used in various embodiments described herein.

[0065] As described herein, controllers are electronic components that manage and coordinate the operation of other components within an electronic device (e.g., controlling inputs, processing data, and/or generating outputs). Examples of controllers can include (i) microcontrollers, including

small, low-power controllers that are commonly used in embedded systems and Internet of Things (IoT) devices; (ii) programmable logic controllers (PLCs) that may be configured to be used in industrial automation systems to control and monitor manufacturing processes; (iii) system-on-a-chip (SoC) controllers that integrate multiple components such as processors, memory, I/O interfaces, and other peripherals into a single chip; and/or DSPs. As described herein, a graphics module is a component or software module that is designed to handle graphical operations and/or processes, and can include a hardware module and/or a software module.

[0066] As described herein, memory refers to electronic components in a computer or electronic device that store data and instructions for the processor to access and manipulate. The devices described herein can include volatile and non-volatile memory. Examples of memory can include (i) random access memory (RAM), such as DRAM, SRAM, DDR RAM or other random access solid state memory devices, configured to store data and instructions temporarily; (ii) read-only memory (ROM) configured to store data and instructions permanently (e.g., one or more portions of system firmware and/or boot loaders); (iii) flash memory, magnetic disk storage devices, optical disk storage devices, other non-volatile solid state storage devices, which can be configured to store data in electronic devices (e.g., universal serial bus (USB) drives, memory cards, and/or solid-state drives (SSDs)); and (iv) cache memory configured to temporarily store frequently accessed data and instructions. Memory, as described herein, can include structured data (e.g., SQL databases, MongoDB databases, GraphQL data, or JSON data). Other examples of memory can include: (i) profile data, including user account data, user settings, and/or other user data stored by the user; (ii) sensor data detected and/or otherwise obtained by one or more sensors; (iii) media content data including stored image data, audio data, documents, and the like; (iv) application data, which can include data collected and/or otherwise obtained and stored during use of an application; and/or any other types of data described herein.

[0067] As described herein, a power system of an electronic device is configured to convert incoming electrical power into a form that can be used to operate the device. A power system can include various components, including (i) a power source, which can be an alternating current (AC) adapter or a direct current (DC) adapter power supply; (ii) a charger input that can be configured to use a wired and/or wireless connection (which may be part of a peripheral interface, such as a USB, micro-USB interface, near-field magnetic coupling, magnetic inductive and magnetic resonance charging, and/or radio frequency (RF) charging); (iii) a power-management integrated circuit, configured to distribute power to various components of the device and ensure that the device operates within safe limits (e.g., regulating voltage, controlling current flow, and/or managing heat dissipation); and/or (iv) a battery configured to store power to provide usable power to components of one or more electronic devices.

[0068] As described herein, peripheral interfaces are electronic components (e.g., of electronic devices) that allow electronic devices to communicate with other devices or peripherals and can provide a means for input and output of data and signals. Examples of peripheral interfaces can include (i) USB and/or micro-USB interfaces configured for connecting devices to an electronic device; (ii) Bluetooth interfaces configured to allow devices to communicate with each other, including Bluetooth low energy (BLE); (iii) near-field communication (NFC) interfaces configured to be short-range wireless interfaces for operations such as access control; (iv) POGO pins, which may be small, spring-loaded pins configured to provide a charging interface; (v) wireless charging interfaces; (vi) global-position system (GPS) interfaces; (vii) Wi-Fi interfaces for providing a connection between a device and a wireless network; and (viii) sensor interfaces.

[0069] As described herein, sensors are electronic components (e.g., in and/or otherwise in electronic communication with electronic devices, such as wearable devices) configured to detect physical and environmental changes and generate electrical signals. Examples of sensors can include (i) imaging sensors for collecting imaging data (e.g., including one or more cameras disposed on a respective electronic device); (ii) biopotential-signal sensors; (iii) inertial

measurement unit (e.g., IMUs) for detecting, for example, angular rate, force, magnetic field, and/or changes in acceleration; (iv) heart rate sensors for measuring a user's heart rate; (v) SpO₂ sensors for measuring blood oxygen saturation and/or other biometric data of a user; (vi) capacitive sensors for detecting changes in potential at a portion of a user's body (e.g., a sensor-skin interface) and/or the proximity of other devices or objects; and (vii) light sensors (e.g., ToF sensors, infrared light sensors, or visible light sensors), and/or sensors for sensing data from the user or the user's environment. As described herein biopotential-signal-sensing components are devices used to measure electrical activity within the body (e.g., biopotential-signal sensors). Some types of biopotential-signal sensors include: (i) electroencephalography (EEG) sensors configured to measure electrical activity in the brain to diagnose neurological disorders; (ii) electrocardiogram sensors configured to measure electrical activity of the heart to diagnose heart problems; (iii) electromyography (EMG) sensors configured to measure the electrical activity of muscles and diagnose neuromuscular disorders; (iv) electrooculography (EOG) sensors configured to measure the electrical activity of eye muscles to detect eye movement and diagnose eye disorders.

[0070] As described herein, an application stored in memory of an electronic device (e.g., software) includes instructions stored in the memory. Examples of such applications include (i) games; (ii) word processors; (iii) messaging applications; (iv) media-streaming applications; (v) financial applications; (vi) calendars; (vii) clocks; (viii) web browsers; (ix) social media applications, (x) camera applications, (xi) web-based applications; (xii) health applications; (xiii) artificial-reality (AR) applications, and/or any other applications that can be stored in memory. The applications can operate in conjunction with data and/or one or more components of a device or communicatively coupled devices to perform one or more operations and/or functions.

[0071] As described herein, communication interface modules can include hardware and/or software capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, or MiWi), custom or standard wired protocols (e.g., Ethernet or HomePlug), and/or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document. A communication interface is a mechanism that enables different systems or devices to exchange information and data with each other, including hardware, software, or a combination of both hardware and software. For example, a communication interface can refer to a physical connector and/or port on a device that enables communication with other devices (e.g., USB, Ethernet, HDMI, or Bluetooth). In some embodiments, a communication interface can refer to a software layer that enables different software programs to communicate with each other (e.g., application programming interfaces (APIs) and protocols such as HTTP and TCP/IP).

[0072] As described herein, a graphics module is a component or software module that is designed to handle graphical operations and/or processes, and can include a hardware module and/or a software module.

[0073] As described herein, non-transitory computer-readable storage media are physical devices or storage medium that can be used to store electronic data in a non-transitory form (e.g., such that the data is stored permanently until it is intentionally deleted or modified).

Example AR Systems

[0074] FIGS. 4A 4B, 4C-1, 4C-2, 4D-1, and 4D-2 illustrate example artificial-reality systems, in accordance with some embodiments. FIG. 4A shows a first AR system **400a** and first example user interactions using a wrist-wearable device **500**, a head-wearable device (e.g., AR device **600**), and/or a handheld intermediary processing device (HIPD) **700**. FIG. 4B shows a second AR system **400b** and second example user interactions using a wrist-wearable device **500**, AR device **600**, and/or an HIPD **700**. FIGS. 4C-1 and 4C-2 show a third AR system **400c** and third example user interactions using a wrist-wearable device **500**, a head-wearable device (e.g., virtual-reality (VR)

device **610**), and/or an HIPD **700**. FIGS. **4D-1** and **4D-2** show a fourth AR system **400d** and fourth example user interactions using a wrist-wearable device **500**, VR device **610**, and/or a smart textile-based garment **800** (e.g., wearable gloves haptic gloves). As the skilled artisan will appreciate upon reading the descriptions provided herein, the above-example AR systems (described in detail below) can perform various functions and/or operations.

[0075] The wrist-wearable device **500** and its constituent components are described below in reference to FIGS. **5A-5B**, the head-wearable devices and their constituent components are described below in reference to FIGS. **6A-6D**, and the HIPD **700** and its constituent components are described below in reference to FIGS. **7A-7B**. The smart textile-based garment **800** and its one or more components are described below in reference to FIGS. **8A-8C**. The wrist-wearable device **500**, the head-wearable devices, and/or the HIPD **700** can communicatively couple via a network **425** (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.). Additionally, the wrist-wearable device **500**, the head-wearable devices, and/or the HIPD **700** can also communicatively couple with one or more servers **430**, computers **440** (e.g., laptops, computers, etc.), mobile devices **450** (e.g., smartphones, tablets, etc.), and/or other electronic devices via the network **425** (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.). Similarly, the smart textile-based garment **800**, when used, can also communicatively couple with the wrist-wearable device **500**, the head-wearable devices, the HIPD **700**, the one or more servers **430**, the computers **440**, the mobile devices **450**, and/or other electronic devices via the network **425**.

[0076] Turning to FIG. **4A**, a user **402** is shown wearing the wrist-wearable device **500** and the AR device **600**, and having the HIPD **700** on their desk. The wrist-wearable device **500**, the AR device **600**, and the HIPD **700** facilitate user interaction with an AR environment. In particular, as shown by the first AR system **400a**, the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** cause presentation of one or more avatars **404**, digital representations of contacts **406**, and virtual objects **408**. As discussed below, the user **402** can interact with the one or more avatars **404**, digital representations of the contacts **406**, and virtual objects **408** via the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700**.

[0077] The user **402** can use any of the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** to provide user inputs. For example, the user **402** can perform one or more hand gestures that are detected by the wrist-wearable device **500** (e.g., using one or more EMG sensors and/or IMUs, described below in reference to FIGS. **5A-5B**) and/or AR device **600** (e.g., using one or more image sensors or cameras, described below in reference to FIGS. **6A-6B**) to provide a user input. Alternatively, or additionally, the user **402** can provide a user input via one or more touch surfaces of the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700**, and/or voice commands captured by a microphone of the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700**. In some embodiments, the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** include a digital assistant to help the user in providing a user input (e.g., completing a sequence of operations, suggesting different operations or commands, providing reminders, confirming a command). In some embodiments, the user **402** can provide a user input via one or more facial gestures and/or facial expressions. For example, cameras of the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** can track the user **402**'s eyes for navigating a user interface.

[0078] The wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** can operate alone or in conjunction to allow the user **402** to interact with the AR environment. In some embodiments, the HIPD **700** is configured to operate as a central hub or control center for the wrist-wearable device **500**, the AR device **600**, and/or another communicatively coupled device. For example, the user **402** can provide an input to interact with the AR environment at any of the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700**, and the HIPD **700** can identify one or more back-end and front-end tasks to cause the performance of the requested interaction and distribute instructions to cause the performance of the one or more back-end and front-end tasks at the wrist-

wearable device **500**, the AR device **600**, and/or the HIPD **700**. In some embodiments, a back-end task is a background-processing task that is not perceptible by the user (e.g., rendering content, decompression, compression, etc.), and a front-end task is a user-facing task that is perceptible to the user (e.g., presenting information to the user, providing feedback to the user, etc.)). As described below in reference to FIGS. 7A-7B, the HIPD **700** can perform the back-end tasks and provide the wrist-wearable device **500** and/or the AR device **600** operational data corresponding to the performed back-end tasks such that the wrist-wearable device **500** and/or the AR device **600** can perform the front-end tasks. In this way, the HIPD **700**, which has more computational resources and greater thermal headroom than the wrist-wearable device **500** and/or the AR device **600**, performs computationally intensive tasks and reduces the computer resource utilization and/or power usage of the wrist-wearable device **500** and/or the AR device **600**.

[0079] In the example shown by the first AR system **400a**, the HIPD **700** identifies one or more back-end tasks and front-end tasks associated with a user request to initiate an AR video call with one or more other users (represented by the avatar **404** and the digital representation of the contact **406**) and distributes instructions to cause the performance of the one or more back-end tasks and front-end tasks. In particular, the HIPD **700** performs back-end tasks for processing and/or rendering image data (and other data) associated with the AR video call and provides operational data associated with the performed back-end tasks to the AR device **600** such that the AR device **600** performs front-end tasks for presenting the AR video call (e.g., presenting the avatar **404** and the digital representation of the contact **406**).

[0080] In some embodiments, the HIPD **700** can operate as a focal or anchor point for causing the presentation of information. This allows the user **402** to be generally aware of where information is presented. For example, as shown in the first AR system **400a**, the avatar **404** and the digital representation of the contact **406** are presented above the HIPD **700**. In particular, the HIPD **700** and the AR device **600** operate in conjunction to determine a location for presenting the avatar **404** and the digital representation of the contact **406**. In some embodiments, information can be presented within a predetermined distance from the HIPD **700** (e.g., within five meters). For example, as shown in the first AR system **400a**, virtual object **408** is presented on the desk some distance from the HIPD **700**. Similar to the above example, the HIPD **700** and the AR device **600** can operate in conjunction to determine a location for presenting the virtual object **408**.

Alternatively, in some embodiments, presentation of information is not bound by the HIPD **700**. More specifically, the avatar **404**, the digital representation of the contact **406**, and the virtual object **408** do not have to be presented within a predetermined distance of the HIPD **700**.

[0081] User inputs provided at the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** are coordinated such that the user can use any device to initiate, continue, and/or complete an operation. For example, the user **402** can provide a user input to the AR device **600** to cause the AR device **600** to present the virtual object **408** and, while the virtual object **408** is presented by the AR device **600**, the user **402** can provide one or more hand gestures via the wrist-wearable device **500** to interact and/or manipulate the virtual object **408**.

[0082] FIG. 4B shows the user **402** wearing the wrist-wearable device **500** and the AR device **600**, and holding the HIPD **700**. In the second AR system **400b**, the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** are used to receive and/or provide one or more messages to a contact of the user **402**. In particular, the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** detect and coordinate one or more user inputs to initiate a messaging application and prepare a response to a received message via the messaging application.

[0083] In some embodiments, the user **402** initiates, via a user input, an application on the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** that causes the application to initiate on at least one device. For example, in the second AR system **400b** the user **402** performs a hand gesture associated with a command for initiating a messaging application (represented by messaging user interface **412**); the wrist-wearable device **500** detects the hand gesture; and, based

on a determination that the user **402** is wearing AR device **600**, causes the AR device **600** to present a messaging user interface **412** of the messaging application. The AR device **600** can present the messaging user interface **412** to the user **402** via its display (e.g., as shown by user **402**'s field of view **410**). In some embodiments, the application is initiated and can be run on the device (e.g., the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700**) that detects the user input to initiate the application, and the device provides another device operational data to cause the presentation of the messaging application. For example, the wrist-wearable device **500** can detect the user input to initiate a messaging application, initiate and run the messaging application, and provide operational data to the AR device **600** and/or the HIPD **700** to cause presentation of the messaging application. Alternatively, the application can be initiated and run at a device other than the device that detected the user input. For example, the wrist-wearable device **500** can detect the hand gesture associated with initiating the messaging application and cause the HIPD **700** to run the messaging application and coordinate the presentation of the messaging application.

[0084] Further, the user **402** can provide a user input provided at the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** to continue and/or complete an operation initiated at another device. For example, after initiating the messaging application via the wrist-wearable device **500** and while the AR device **600** presents the messaging user interface **412**, the user **402** can provide an input at the HIPD **700** to prepare a response (e.g., shown by the swipe gesture performed on the HIPD **700**). The user **402**'s gestures performed on the HIPD **700** can be provided and/or displayed on another device. For example, the user **402**'s swipe gestures performed on the HIPD **700** are displayed on a virtual keyboard of the messaging user interface **412** displayed by the AR device **600**.

[0085] In some embodiments, the wrist-wearable device **500**, the AR device **600**, the HIPD **700**, and/or other communicatively coupled devices can present one or more notifications to the user **402**. The notification can be an indication of a new message, an incoming call, an application update, a status update, etc. The user **402** can select the notification via the wrist-wearable device **500**, the AR device **600**, or the HIPD **700** and cause presentation of an application or operation associated with the notification on at least one device. For example, the user **402** can receive a notification that a message was received at the wrist-wearable device **500**, the AR device **600**, the HIPD **700**, and/or other communicatively coupled device and provide a user input at the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** to review the notification, and the device detecting the user input can cause an application associated with the notification to be initiated and/or presented at the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700**.

[0086] While the above example describes coordinated inputs used to interact with a messaging application, the skilled artisan will appreciate upon reading the descriptions that user inputs can be coordinated to interact with any number of applications including, but not limited to, gaming applications, social media applications, camera applications, web-based applications, financial applications, etc. For example, the AR device **600** can present to the user **402** game application data and the HIPD **700** can use a controller to provide inputs to the game. Similarly, the user **402** can use the wrist-wearable device **500** to initiate a camera of the AR device **600**, and the user can use the wrist-wearable device **500**, the AR device **600**, and/or the HIPD **700** to manipulate the image capture (e.g., zoom in or out, apply filters, etc.) and capture image data.

[0087] Turning to FIGS. **4C-1** and **4C-2**, the user **402** is shown wearing the wrist-wearable device **500** and a VR device **610**, and holding the HIPD **700**. In the third AR system **400c**, the wrist-wearable device **500**, the VR device **610**, and/or the HIPD **700** are used to interact within an AR environment, such as a VR game or other AR application. While the VR device **610** present a representation of a VR game (e.g., first AR game environment **420**) to the user **402**, the wrist-wearable device **500**, the VR device **610**, and/or the HIPD **700** detect and coordinate one or more user inputs to allow the user **402** to interact with the VR game.

[0088] In some embodiments, the user **402** can provide a user input via the wrist-wearable device **500**, the VR device **610**, and/or the HIPD **700** that causes an action in a corresponding AR environment. For example, the user **402** in the third AR system **400c** (shown in FIG. **4C-1**) raises the HIPD **700** to prepare for a swing in the first AR game environment **420**. The VR device **610**, responsive to the user **402** raising the HIPD **700**, causes the AR representation of the user **422** to perform a similar action (e.g., raise a virtual object, such as a virtual sword **424**). In some embodiments, each device uses respective sensor data and/or image data to detect the user input and provide an accurate representation of the user **402**'s motion. For example, image sensors **758** (e.g., SLAM cameras or other cameras discussed below in FIGS. **7A** and **7B**) of the HIPD **700** can be used to detect a position of the **700** relative to the user **402**'s body such that the virtual object can be positioned appropriately within the first AR game environment **420**; sensor data from the wrist-wearable device **500** can be used to detect a velocity at which the user **402** raises the HIPD **700** such that the AR representation of the user **422** and the virtual sword **424** are synchronized with the user **402**'s movements; and image sensors **626** (FIGS. **6A-6C**) of the VR device **610** can be used to represent the user **402**'s body, boundary conditions, or real-world objects within the first AR game environment **420**.

[0089] In FIG. **4C-2**, the user **402** performs a downward swing while holding the HIPD **700**. The user **402**'s downward swing is detected by the wrist-wearable device **500**, the VR device **610**, and/or the HIPD **700** and a corresponding action is performed in the first AR game environment **420**. In some embodiments, the data captured by each device is used to improve the user's experience within the AR environment. For example, sensor data of the wrist-wearable device **500** can be used to determine a speed and/or force at which the downward swing is performed and image sensors of the HIPD **700** and/or the VR device **610** can be used to determine a location of the swing and how it should be represented in the first AR game environment **420**, which, in turn, can be used as inputs for the AR environment (e.g., game mechanics, which can use detected speed, force, locations, and/or aspects of the user **402**'s actions to classify a user's inputs (e.g., user performs a light strike, hard strike, critical strike, glancing strike, miss) or calculate an output (e.g., amount of damage)).

[0090] While the wrist-wearable device **500**, the VR device **610**, and/or the HIPD **700** are described as detecting user inputs, in some embodiments, user inputs are detected at a single device (with the single device being responsible for distributing signals to the other devices for performing the user input). For example, the HIPD **700** can operate an application for generating the first AR game environment **420** and provide the VR device **610** with corresponding data for causing the presentation of the first AR game environment **420**, as well as detect the **402**'s movements (while holding the HIPD **700**) to cause the performance of corresponding actions within the first AR game environment **420**. Additionally or alternatively, in some embodiments, operational data (e.g., sensor data, image data, application data, device data, and/or other data) of one or more devices is provide to a single device (e.g., the HIPD **700**) to process the operational data and cause respective devices to perform an action associated with processed operational data.

[0091] FIGS. **4D-1** and **4D-2**, the user **402** is shown wearing the wrist-wearable device **500**, the VR device **610**, smart textile-based garments **800**. In the fourth AR system **400d**, the wrist-wearable device **500**, the VR device **610**, and/or the smart textile-based garments **800** are used to interact within an AR environment (e.g., any AR system described above in reference to FIGS. **4A-4C-2**). While the VR device **610** present a representation of a VR game (e.g., second AR game environment **430**) to the user **402**, the wrist-wearable device **500**, the VR device **610**, and/or the smart textile-based garments **800** detect and coordinate one or more user inputs to allow the user **402** to interact with the AR environment.

[0092] In some embodiments, the user **402** can provide a user input via the wrist-wearable device **500**, the VR device **610**, and/or the smart textile-based garments **800** that causes an action in a corresponding AR environment. For example, the user **402** in the fourth AR system **400d** (shown in

FIG. 4D-1) raises a hand wearing the smart textile-based garments **800** to prepare for cast spell or throw an object within the second AR game environment **430**. The VR device **610**, responsive to the user **402** holding up their hand (wearing a smart textile-based garments **800**), causes the AR representation of the user **422** to perform a similar action (e.g., hold a virtual object, such as a casting a fireball **434**). In some embodiments, each device uses respective sensor data and/or image data to detect the user input and provide an accurate representation of the user **402**'s motion.

[0093] In FIG. 4D-2, the user **402** performs a throwing motion while wearing the smart textile-based garment **800**. The user **402**'s throwing motion is detected by the wrist-wearable device **500**, the VR device **610**, and/or the smart textile-based garments **800** and a corresponding action is performed in the second AR game environment **430**. As described above, the data captured by each device is used to improve the user's experience within the AR environment. Although not shown, the smart textile-based garments **800** can be used in conjunction with an AR device **610** and/or an HIPD **700**.

[0094] Having discussed example AR systems, devices for interacting with such AR systems, and other computing systems more generally, will now be discussed in greater detail below. Some definitions of devices and components that can be included in some or all of the example devices discussed below are defined here for ease of reference. A skilled artisan will appreciate that certain types of the components described below may be more suitable for a particular set of devices, and less suitable for a different set of devices. But subsequent reference to the components defined here should be considered to be encompassed by the definitions provided.

[0095] In some embodiments discussed below example devices and systems, including electronic devices and systems, will be discussed. Such example devices and systems are not intended to be limiting, and one of skill in the art will understand that alternative devices and systems to the example devices and systems described herein may be used to perform the operations and construct the systems and device that are described herein.

[0096] As described herein, an electronic device is a device that uses electrical energy to perform a specific function. It can be any physical object that contains electronic components such as transistors, resistors, capacitors, diodes, and integrated circuits. Examples of electronic devices include smartphones, laptops, digital cameras, televisions, gaming consoles, and music players, as well as the example electronic devices discussed herein. As described herein, an intermediary electronic device is a device that sits between two other electronic devices, and/or a subset of components of one or more electronic devices and facilitates communication, and/or data processing and/or data transfer between the respective electronic devices and/or electronic components.

Example Wrist-Wearable Devices

[0097] FIGS. 5A and 5B illustrate an example wrist-wearable device **500**, in accordance with some embodiments. FIG. 5A illustrates components of the wrist-wearable device **500**, which can be used individually or in combination, including combinations that include other electronic devices and/or electronic components.

[0098] FIG. 5A shows a wearable band **510** and a watch body **520** (or capsule) being coupled, as discussed below, to form the wrist-wearable device **500**. The wrist-wearable device **500** can perform various functions and/or operations associated with navigating through user interfaces and selectively opening applications.

[0099] As will be described in more detail below, operations executed by the wrist-wearable device **500** can include (i) presenting content to a user (e.g., displaying visual content via a display **505**); (ii) detecting (e.g., sensing) user input (e.g., sensing a touch on peripheral button **523** and/or at a touch screen of the display **505**, a hand gesture detected by sensors (e.g., biopotential sensors)); (iii) sensing biometric data via one or more sensors **513** (e.g., neuromuscular signals, heart rate, temperature, sleep, etc.); messaging (e.g., text, speech, video, etc.); image capture via one or more imaging devices or cameras **525**; wireless communications (e.g., cellular, near field, Wi-Fi,

personal area network, etc.); location determination; financial transactions; providing haptic feedback; alarms; notifications; biometric authentication; health monitoring; sleep monitoring. [0100] The above-example functions can be executed independently in the watch body **520**, independently in the wearable band **510**, and/or via an electronic communication between the watch body **520** and the wearable band **510**. In some embodiments, functions can be executed on the wrist-wearable device **500** while an AR environment is being presented (e.g., via one of the AR systems **400a** to **400d**). As the skilled artisan will appreciate upon reading the descriptions provided herein, the novel wearable devices described herein can be used with other types of AR environments.

[0101] The wearable band **510** can be configured to be worn by a user such that an inner (or inside) surface of the wearable structure **511** of the wearable band **510** is in contact with the user's skin. When worn by a user, sensors **513** contact the user's skin. The sensors **513** can sense biometric data such as a user's heart rate, saturated oxygen level, temperature, sweat level, neuromuscular signal sensors, or a combination thereof. The sensors **513** can also sense data about a user's environment, including a user's motion, altitude, location, orientation, gait, acceleration, position, or a combination thereof. In some embodiments, the sensors **513** are configured to track a position and/or motion of the wearable band **510**. The one or more sensors **513** can include any of the sensors defined above and/or discussed below with respect to FIG. 5B.

[0102] The one or more sensors **513** can be distributed on an inside and/or an outside surface of the wearable band **510**. In some embodiments, the one or more sensors **513** are uniformly spaced along the wearable band **510**. Alternatively, in some embodiments, the one or more sensors **513** are positioned at distinct points along the wearable band **510**. As shown in FIG. 5A, the one or more sensors **513** can be the same or distinct. For example, in some embodiments, the one or more sensors **513** can be shaped as a pill (e.g., sensor **513a**), an oval, a circle a square, an oblong (e.g., sensor **513c**) and/or any other shape that maintains contact with the user's skin (e.g., such that neuromuscular signal and/or other biometric data can be accurately measured at the user's skin). In some embodiments, the one or more sensors **513** are aligned to form pairs of sensors (e.g., for sensing neuromuscular signals based on differential sensing within each respective sensor). For example, sensor **513b** is aligned with an adjacent sensor to form sensor pair **514a** and sensor **513d** is aligned with an adjacent sensor to form sensor pair **514b**. In some embodiments, the wearable band **510** does not have a sensor pair. Alternatively, in some embodiments, the wearable band **510** has a predetermined number of sensor pairs (one pair of sensors, three pairs of sensors, four pairs of sensors, six pairs of sensors, sixteen pairs of sensors, etc.).

[0103] The wearable band **510** can include any suitable number of sensors **513**. In some embodiments, the number and arrangements of sensors **513** depend on the particular application for which the wearable band **510** is used. For instance, a wearable band **510** configured as an armband, wristband, or chest-band may include a plurality of sensors **513** with different number of sensors **513** and different arrangement for each use case, such as medical use cases, compared to gaming or general day-to-day use cases.

[0104] In accordance with some embodiments, the wearable band **510** further includes an electrical ground electrode and a shielding electrode. The electrical ground and shielding electrodes, like the sensors **513**, can be distributed on the inside surface of the wearable band **510** such that they contact a portion of the user's skin. For example, the electrical ground and shielding electrodes can be at an inside surface of coupling mechanism **516** or an inside surface of a wearable structure **511**. The electrical ground and shielding electrodes can be formed and/or use the same components as the sensors **513**. In some embodiments, the wearable band **510** includes more than one electrical ground electrode and more than one shielding electrode.

[0105] The sensors **513** can be formed as part of the wearable structure **511** of the wearable band **510**. In some embodiments, the sensors **513** are flush or substantially flush with the wearable structure **511** such that they do not extend beyond the surface of the wearable structure **511**. While

flush with the wearable structure **511**, the sensors **513** are still configured to contact the user's skin (e.g., via a skin-contacting surface). Alternatively, in some embodiments, the sensors **513** extend beyond the wearable structure **511** a predetermined distance (e.g., 0.1 mm to 2 mm) to make contact and depress into the user's skin. In some embodiments, the sensors **513** are coupled to an actuator (not shown) configured to adjust an extension height (e.g., a distance from the surface of the wearable structure **511**) of the sensors **513** such that the sensors **513** make contact and depress into the user's skin. In some embodiments, the actuators adjust the extension height between 0.01 mm to 1.2 mm. This allows the user to customize the positioning of the sensors **513** to improve the overall comfort of the wearable band **510** when worn while still allowing the sensors **513** to contact the user's skin. In some embodiments, the sensors **513** are indistinguishable from the wearable structure **511** when worn by the user.

[0106] The wearable structure **511** can be formed of an elastic material, elastomers, etc., configured to be stretched and fitted to be worn by the user. In some embodiments, the wearable structure **511** is a textile or woven fabric. As described above, the sensors **513** can be formed as part of a wearable structure **511**. For example, the sensors **513** can be molded into the wearable structure **511** or be integrated into a woven fabric (e.g., the sensors **513** can be sewn into the fabric and mimic the pliability of fabric (e.g., the sensors **513** can be constructed from a series of woven strands of fabric)).

[0107] The wearable structure **511** can include flexible electronic connectors that interconnect the sensors **513**, the electronic circuitry, and/or other electronic components (described below in reference to FIG. 5B) that are enclosed in the wearable band **510**. In some embodiments, the flexible electronic connectors are configured to interconnect the sensors **513**, the electronic circuitry, and/or other electronic components of the wearable band **510** with respective sensors and/or other electronic components of another electronic device (e.g., watch body **520**). The flexible electronic connectors are configured to move with the wearable structure **511** such that the user adjustment to the wearable structure **511** (e.g., resizing, pulling, folding, etc.) does not stress or strain the electrical coupling of components of the wearable band **510**.

[0108] As described above, the wearable band **510** is configured to be worn by a user. In particular, the wearable band **510** can be shaped or otherwise manipulated to be worn by a user. For example, the wearable band **510** can be shaped to have a substantially circular shape such that it can be configured to be worn on the user's lower arm or wrist. Alternatively, the wearable band **510** can be shaped to be worn on another body part of the user, such as the user's upper arm (e.g., around a bicep), forearm, chest, legs, etc. The wearable band **510** can include a retaining mechanism **512** (e.g., a buckle, a hook and loop fastener, etc.) for securing the wearable band **510** to the user's wrist or other body part. While the wearable band **510** is worn by the user, the sensors **513** sense data (referred to as sensor data) from the user's skin. In particular, the sensors **513** of the wearable band **510** obtain (e.g., sense and record) neuromuscular signals.

[0109] The sensed data (e.g., sensed neuromuscular signals) can be used to detect and/or determine the user's intention to perform certain motor actions. In particular, the sensors **513** sense and record neuromuscular signals from the user as the user performs muscular activations (e.g., movements, gestures, etc.). The detected and/or determined motor actions (e.g., phalange (or digits) movements, wrist movements, hand movements, and/or other muscle intentions) can be used to determine control commands or control information (instructions to perform certain commands after the data is sensed) for causing a computing device to perform one or more input commands. For example, the sensed neuromuscular signals can be used to control certain user interfaces displayed on the display **505** of the wrist-wearable device **500** and/or can be transmitted to a device responsible for rendering an artificial-reality environment (e.g., a head-mounted display) to perform an action in an associated artificial-reality environment, such as to control the motion of a virtual device displayed to the user. The muscular activations performed by the user can include static gestures, such as placing the user's hand palm down on a table; dynamic gestures, such as grasping a physical or

virtual object; and covert gestures that are imperceptible to another person, such as slightly tensing a joint by co-contracting opposing muscles or using sub-muscular activations. The muscular activations performed by the user can include symbolic gestures (e.g., gestures mapped to other gestures, interactions, or commands, for example, based on a gesture vocabulary that specifies the mapping of gestures to commands).

[0110] The sensor data sensed by the sensors **513** can be used to provide a user with an enhanced interaction with a physical object (e.g., devices communicatively coupled with the wearable band **510**) and/or a virtual object in an artificial-reality application generated by an artificial-reality system (e.g., user interface objects presented on the display **505** or another computing device (e.g., a smartphone)).

[0111] In some embodiments, the wearable band **510** includes one or more haptic devices **546** (FIG. **5B**; e.g., a vibratory haptic actuator) that are configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user's skin. The sensors **513**, and/or the haptic devices **546** can be configured to operate in conjunction with multiple applications including, without limitation, health monitoring, social media, games, and artificial reality (e.g., the applications associated with artificial reality).

[0112] The wearable band **510** can also include coupling mechanism **516** (e.g., a cradle or a shape of the coupling mechanism can correspond to shape of the watch body **520** of the wrist-wearable device **500**) for detachably coupling a capsule (e.g., a computing unit) or watch body **520** (via a coupling surface of the watch body **520**) to the wearable band **510**. In particular, the coupling mechanism **516** can be configured to receive a coupling surface proximate to the bottom side of the watch body **520** (e.g., a side opposite to a front side of the watch body **520** where the display **505** is located), such that a user can push the watch body **520** downward into the coupling mechanism **516** to attach the watch body **520** to the coupling mechanism **516**. In some embodiments, the coupling mechanism **516** can be configured to receive a top side of the watch body **520** (e.g., a side proximate to the front side of the watch body **520** where the display **505** is located) that is pushed upward into the cradle, as opposed to being pushed downward into the coupling mechanism **516**. In some embodiments, the coupling mechanism **516** is an integrated component of the wearable band **510** such that the wearable band **510** and the coupling mechanism **516** are a single unitary structure. In some embodiments, the coupling mechanism **516** is a type of frame or shell that allows the watch body **520** coupling surface to be retained within or on the wearable band **510** coupling mechanism **516** (e.g., a cradle, a tracker band, a support base, a clasp, etc.).

[0113] The coupling mechanism **516** can allow for the watch body **520** to be detachably coupled to the wearable band **510** through a friction fit, magnetic coupling, a rotation-based connector, a shear-pin coupler, a retention spring, one or more magnets, a clip, a pin shaft, a hook and loop fastener, or a combination thereof. A user can perform any type of motion to couple the watch body **520** to the wearable band **510** and to decouple the watch body **520** from the wearable band **510**. For example, a user can twist, slide, turn, push, pull, or rotate the watch body **520** relative to the wearable band **510**, or a combination thereof, to attach the watch body **520** to the wearable band **510** and to detach the watch body **520** from the wearable band **510**. Alternatively, as discussed below, in some embodiments, the watch body **520** can be decoupled from the wearable band **510** by actuation of the release mechanism **529**.

[0114] The wearable band **510** can be coupled with a watch body **520** to increase the functionality of the wearable band **510** (e.g., converting the wearable band **510** into a wrist-wearable device **500**, adding an additional computing unit and/or battery to increase computational resources and/or a battery life of the wearable band **510**, adding additional sensors to improve sensed data, etc.). As described above, the wearable band **510** (and the coupling mechanism **516**) is configured to operate independently (e.g., execute functions independently) from watch body **520**. For example, the coupling mechanism **516** can include one or more sensors **513** that contact a user's skin when the wearable band **510** is worn by the user and provide sensor data for determining control commands.

[0115] A user can detach the watch body **520** (or capsule) from the wearable band **510** in order to reduce the encumbrance of the wrist-wearable device **500** to the user. For embodiments in which the watch body **520** is removable, the watch body **520** can be referred to as a removable structure, such that in these embodiments the wrist-wearable device **500** includes a wearable portion (e.g., the wearable band **510**) and a removable structure (the watch body **520**).

[0116] Turning to the watch body **520**, the watch body **520** can have a substantially rectangular or circular shape. The watch body **520** is configured to be worn by the user on their wrist or on another body part. More specifically, the watch body **520** is sized to be easily carried by the user, attached on a portion of the user's clothing, and/or coupled to the wearable band **510** (forming the wrist-wearable device **500**). As described above, the watch body **520** can have a shape corresponding to the coupling mechanism **516** of the wearable band **510**. In some embodiments, the watch body **520** includes a single release mechanism **529** or multiple release mechanisms (e.g., two release mechanisms **529** positioned on opposing sides of the watch body **520**, such as spring-loaded buttons) for decoupling the watch body **520** and the wearable band **510**. The release mechanism **529** can include, without limitation, a button, a knob, a plunger, a handle, a lever, a fastener, a clasp, a dial, a latch, or a combination thereof.

[0117] A user can actuate the release mechanism **529** by pushing, turning, lifting, depressing, shifting, or performing other actions on the release mechanism **529**. Actuation of the release mechanism **529** can release (e.g., decouple) the watch body **520** from the coupling mechanism **516** of the wearable band **510**, allowing the user to use the watch body **520** independently from wearable band **510**, and vice versa. For example, decoupling the watch body **520** from the wearable band **510** can allow the user to capture images using rear-facing camera **525B**. Although the coupling mechanism **516** is shown positioned at a corner of watch body **520**, the release mechanism **529** can be positioned anywhere on watch body **520** that is convenient for the user to actuate. In addition, in some embodiments, the wearable band **510** can also include a respective release mechanism for decoupling the watch body **520** from the coupling mechanism **516**. In some embodiments, the release mechanism **529** is optional and the watch body **520** can be decoupled from the coupling mechanism **516** as described above (e.g., via twisting, rotating, etc.).

[0118] The watch body **520** can include one or more peripheral buttons **523** and **527** for performing various operations at the watch body **520**. For example, the peripheral buttons **523** and **527** can be used to turn on or wake (e.g., transition from a sleep state to an active state) the display **505**, unlock the watch body **520**, increase or decrease a volume, increase or decrease brightness, interact with one or more applications, interact with one or more user interfaces, etc. Additionally, or alternatively, in some embodiments, the display **505** operates as a touch screen and allows the user to provide one or more inputs for interacting with the watch body **520**.

[0119] In some embodiments, the watch body **520** includes one or more sensors **521**. The sensors **521** of the watch body **520** can be the same or distinct from the sensors **513** of the wearable band **510**. The sensors **521** of the watch body **520** can be distributed on an inside and/or an outside surface of the watch body **520**. In some embodiments, the sensors **521** are configured to contact a user's skin when the watch body **520** is worn by the user. For example, the sensors **521** can be placed on the bottom side of the watch body **520** and the coupling mechanism **516** can be a cradle with an opening that allows the bottom side of the watch body **520** to directly contact the user's skin. Alternatively, in some embodiments, the watch body **520** does not include sensors that are configured to contact the user's skin (e.g., including sensors internal and/or external to the watch body **520** that configured to sense data of the watch body **520** and the watch body **520**'s surrounding environment). In some embodiments, the sensors **513** are configured to track a position and/or motion of the watch body **520**.

[0120] The watch body **520** and the wearable band **510** can share data using a wired communication method (e.g., a Universal Asynchronous Receiver/Transmitter (UART), a USB transceiver, etc.) and/or a wireless communication method (e.g., near field communication,

Bluetooth, etc.). For example, the watch body **520** and the wearable band **510** can share data sensed by the sensors **513** and **521**, as well as application- and device-specific information (e.g., active and/or available applications), output devices (e.g., display, speakers, etc.), input devices (e.g., touch screen, microphone, imaging sensors, etc.).

[0121] In some embodiments, the watch body **520** can include, without limitation, a front-facing camera **525A** and/or a rear-facing camera **525B**, sensors **521** (e.g., a biometric sensor, an IMU sensor, a heart rate sensor, a saturated oxygen sensor, a neuromuscular signal sensor, an altimeter sensor, a temperature sensor, a bioimpedance sensor, a pedometer sensor, an optical sensor (e.g., imaging sensor **563**; FIG. 5B), a touch sensor, a sweat sensor, etc.). In some embodiments, the watch body **520** can include one or more haptic devices **576** (FIG. 5B; a vibratory haptic actuator) that is configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user. The sensors **521** and/or the haptic device **576** can also be configured to operate in conjunction with multiple applications including, without limitation, health-monitoring applications, social media applications, game applications, and artificial-reality applications (e.g., the applications associated with artificial reality).

[0122] As described above, the watch body **520** and the wearable band **510**, when coupled, can form the wrist-wearable device **500**. When coupled, the watch body **520** and wearable band **510** operate as a single device to execute functions (operations, detections, communications, etc.) described herein. In some embodiments, each device is provided with particular instructions for performing the one or more operations of the wrist-wearable device **500**. For example, in accordance with a determination that the watch body **520** does not include neuromuscular signal sensors, the wearable band **510** can include alternative instructions for performing associated instructions (e.g., providing sensed neuromuscular signal data to the watch body **520** via a different electronic device). Operations of the wrist-wearable device **500** can be performed by the watch body **520** alone or in conjunction with the wearable band **510** (e.g., via respective processors and/or hardware components) and vice versa. In some embodiments, operations of the wrist-wearable device **500**, the watch body **520**, and/or the wearable band **510** can be performed in conjunction with one or more processors and/or hardware components of another communicatively coupled device (e.g., the HIPD **700**; FIGS. 7A-7B).

[0123] As described below with reference to the block diagram of FIG. 5B, the wearable band **510** and/or the watch body **520** can each include independent resources required to independently execute functions. For example, the wearable band **510** and/or the watch body **520** can each include a power source (e.g., a battery), a memory, data storage, a processor (e.g., a central processing unit (CPU)), communications, a light source, and/or input/output devices.

[0124] FIG. 5B shows block diagrams of a computing system **530** corresponding to the wearable band **510**, and a computing system **560** corresponding to the watch body **520**, according to some embodiments. A computing system of the wrist-wearable device **500** includes a combination of components of the wearable band computing system **530** and the watch body computing system **560**, in accordance with some embodiments.

[0125] The watch body **520** and/or the wearable band **510** can include one or more components shown in watch body computing system **560**. In some embodiments, a single integrated circuit includes all or a substantial portion of the components of the watch body computing system **560** are included in a single integrated circuit. Alternatively, in some embodiments, components of the watch body computing system **560** are included in a plurality of integrated circuits that are communicatively coupled. In some embodiments, the watch body computing system **560** is configured to couple (e.g., via a wired or wireless connection) with the wearable band computing system **530**, which allows the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0126] The watch body computing system **560** can include one or more processors **579**, a controller **577**, a peripherals interface **561**, a power system **595**, and memory (e.g., a memory **580**), each of

which are defined above and described in more detail below.

[0127] The power system **595** can include a charger input **596**, a power-management integrated circuit (PMIC) **597**, and a battery **598**, each of which are defined above. In some embodiments, a watch body **520** and a wearable band **510** can have respective charger inputs (e.g., charger input **596** and **557**), respective batteries (e.g., battery **598** and **559**), and can share power with each other (e.g., the watch body **520** can power and/or charge the wearable band **510**, and vice versa). Although watch body **520** and/or the wearable band **510** can include respective charger inputs, a single charger input can charge both devices when coupled. The watch body **520** and the wearable band **510** can receive a charge using a variety of techniques. In some embodiments, the watch body **520** and the wearable band **510** can use a wired charging assembly (e.g., power cords) to receive the charge. Alternatively, or in addition, the watch body **520** and/or the wearable band **510** can be configured for wireless charging. For example, a portable charging device can be designed to mate with a portion of watch body **520** and/or wearable band **510** and wirelessly deliver usable power to a battery of watch body **520** and/or wearable band **510**. The watch body **520** and the wearable band **510** can have independent power systems (e.g., power system **595** and **556**) to enable each to operate independently. The watch body **520** and wearable band **510** can also share power (e.g., one can charge the other) via respective PMICs (e.g., PMICs **597** and **558**) that can share power over power and ground conductors and/or over wireless charging antennas.

[0128] In some embodiments, the peripherals interface **561** can include one or more sensors **521**, many of which listed below are defined above. The sensors **521** can include one or more coupling sensors **562** for detecting when the watch body **520** is coupled with another electronic device (e.g., a wearable band **510**). The sensors **521** can include imaging sensors **563** (one or more of the cameras **525** and/or separate imaging sensors **563** (e.g., thermal-imaging sensors)). In some embodiments, the sensors **521** include one or more SpO₂ sensors **564**. In some embodiments, the sensors **521** include one or more biopotential-signal sensors (e.g., EMG sensors **565**, which may be disposed on a user-facing portion of the watch body **520** and/or the wearable band **510**). In some embodiments, the sensors **521** include one or more capacitive sensors **566**. In some embodiments, the sensors **521** include one or more heart rate sensors **567**. In some embodiments, the sensors **521** include one or more IMUs **568**. In some embodiments, one or more IMUs **568** can be configured to detect movement of a user's hand or other location that the watch body **520** is placed or held.

[0129] In some embodiments, the peripherals interface **561** includes an NFC component **569**, a global-position system (GPS) component **570**, a long-term evolution (LTE) component **571**, and/or a Wi-Fi and/or Bluetooth communication component **572**. In some embodiments, the peripherals interface **561** includes one or more buttons **573** (e.g., the peripheral buttons **523** and **527** in FIG. 5A), which, when selected by a user, cause operations to be performed at the watch body **520**. In some embodiments, the peripherals interface **561** includes one or more indicators, such as a light emitting diode (LED), to provide a user with visual indicators (e.g., message received, low battery, an active microphone, and/or a camera, etc.).

[0130] The watch body **520** can include at least one display **505** for displaying visual representations of information or data to the user, including user-interface elements and/or three-dimensional (3D) virtual objects. The display can also include a touch screen for inputting user inputs, such as touch gestures, swipe gestures, and the like. The watch body **520** can include at least one speaker **574** and at least one microphone **575** for providing audio signals to the user and receiving audio input from the user. The user can provide user inputs through the microphone **575** and can also receive audio output from the speaker **574** as part of a haptic event provided by the haptic controller **578**. The watch body **520** can include at least one camera **525**, including a front-facing camera **525A** and a rear-facing camera **525B**. The cameras **525** can include ultra-wide-angle cameras, wide-angle cameras, fish-eye cameras, spherical cameras, telephoto cameras, a depth-sensing cameras, or other types of cameras.

[0131] The watch body computing system **560** can include one or more haptic controllers **578** and

associated componentry (e.g., haptic devices **576**) for providing haptic events at the watch body **520** (e.g., a vibrating sensation or audio output in response to an event at the watch body **520**). The haptic controllers **578** can communicate with one or more haptic devices **576**, such as electroacoustic devices, including a speaker of the one or more speakers **574** and/or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). The haptic controller **578** can provide haptic events to respective haptic actuators that are capable of being sensed by a user of the watch body **520**. In some embodiments, the one or more haptic controllers **578** can receive input signals from an application of the applications **582**. [0132] In some embodiments, the computer system **530** and/or the computer system **560** can include memory **580**, which can be controlled by a memory controller of the one or more controllers **577** and/or one or more processors **579**. In some embodiments, software components stored in the memory **580** include one or more applications **582** configured to perform operations at the watch body **520**. In some embodiments, the one or more applications **582** include games, word processors, messaging applications, calling applications, web browsers, social media applications, media streaming applications, financial applications, calendars, clocks, etc. In some embodiments, software components stored in the memory **580** include one or more communication interface modules **583** as defined above. In some embodiments, software components stored in the memory **580** include one or more graphics modules **584** for rendering, encoding, and/or decoding audio and/or visual data; and one or more data management modules **585** for collecting, organizing, and/or providing access to the data **587** stored in memory **580**. In some embodiments, one or more of applications **582** and/or one or more modules can work in conjunction with one another to perform various tasks at the watch body **520**.

[0133] In some embodiments, software components stored in the memory **580** can include one or more operating systems **581** (e.g., a Linux-based operating system, an Android operating system, etc.). The memory **580** can also include data **587**. The data **587** can include profile data **588A**, sensor data **589A**, media content data **590**, and application data **591**.

[0134] It should be appreciated that the watch body computing system **560** is an example of a computing system within the watch body **520**, and that the watch body **520** can have more or fewer components than shown in the watch body computing system **560**, combine two or more components, and/or have a different configuration and/or arrangement of the components. The various components shown in watch body computing system **560** are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application-specific integrated circuits.

[0135] Turning to the wearable band computing system **530**, one or more components that can be included in the wearable band **510** are shown. The wearable band computing system **530** can include more or fewer components than shown in the watch body computing system **560**, combine two or more components, and/or have a different configuration and/or arrangement of some or all of the components. In some embodiments, all, or a substantial portion of the components of the wearable band computing system **530** are included in a single integrated circuit. Alternatively, in some embodiments, components of the wearable band computing system **530** are included in a plurality of integrated circuits that are communicatively coupled. As described above, in some embodiments, the wearable band computing system **530** is configured to couple (e.g., via a wired or wireless connection) with the watch body computing system **560**, which allows the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0136] The wearable band computing system **530**, similar to the watch body computing system **560**, can include one or more processors **549**, one or more controllers **547** (including one or more haptics controller **548**), a peripherals interface **531** that can include one or more sensors **513** and

other peripheral devices, power source (e.g., a power system **556**), and memory (e.g., a memory **550**) that includes an operating system (e.g., an operating system **551**), data (e.g., data **554** including profile data **588B**, sensor data **589B**, etc.), and one or more modules (e.g., a communications interface module **552**, a data management module **553**, etc.).

[0137] The one or more sensors **513** can be analogous to sensors **521** of the computer system **560** in light of the definitions above. For example, sensors **513** can include one or more coupling sensors **532**, one or more SpO2 sensors **534**, one or more EMG sensors **535**, one or more capacitive sensors **536**, one or more heart rate sensors **537**, and one or more IMU sensors **538**.

[0138] The peripherals interface **531** can also include other components analogous to those included in the peripheral interface **561** of the computer system **560**, including an NFC component **539**, a GPS component **540**, an LTE component **541**, a Wi-Fi and/or Bluetooth communication component **542**, and/or one or more haptic devices **576** as described above in reference to peripherals interface **561**. In some embodiments, the peripherals interface **531** includes one or more buttons **543**, a display **533**, a speaker **544**, a microphone **545**, and a camera **555**. In some embodiments, the peripherals interface **531** includes one or more indicators, such as an LED.

[0139] It should be appreciated that the wearable band computing system **530** is an example of a computing system within the wearable band **510**, and that the wearable band **510** can have more or fewer components than shown in the wearable band computing system **530**, combine two or more components, and/or have a different configuration and/or arrangement of the components. The various components shown in wearable band computing system **530** can be implemented in one or a combination of hardware, software, and firmware, including one or more signal processing and/or application-specific integrated circuits.

[0140] The wrist-wearable device **500** with respect to FIG. 5A is an example of the wearable band **510** and the watch body **520** coupled, so the wrist-wearable device **500** will be understood to include the components shown and described for the wearable band computing system **530** and the watch body computing system **560**. In some embodiments, wrist-wearable device **500** has a split architecture (e.g., a split mechanical architecture or a split electrical architecture) between the watch body **520** and the wearable band **510**. In other words, all of the components shown in the wearable band computing system **530** and the watch body computing system **560** can be housed or otherwise disposed in a combined watch device **500**, or within individual components of the watch body **520**, wearable band **510**, and/or portions thereof (e.g., a coupling mechanism **516** of the wearable band **510**).

[0141] The techniques described above can be used with any device for sensing neuromuscular signals, including the arm-wearable devices of FIG. 5A-5B, but could also be used with other types of wearable devices for sensing neuromuscular signals (such as body-wearable or head-wearable devices that might have neuromuscular sensors closer to the brain or spinal column).

[0142] In some embodiments, a wrist-wearable device **500** can be used in conjunction with a head-wearable device described below (e.g., AR device **600** and VR device **610**) and/or an HIPD **700**, and the wrist-wearable device **500** can also be configured to be used to allow a user to control aspect of the artificial reality (e.g., by using EMG-based gestures to control user interface objects in the artificial reality and/or by allowing a user to interact with the touchscreen on the wrist-wearable device to also control aspects of the artificial reality). In some embodiments, a wrist-wearable device **500** can also be used in conjunction with a wearable garment, such as smart textile-based garment **800** described below in reference to FIGS. 8A-8C. Having thus described example wrist-wearable device, attention will now be turned to example head-wearable devices, such AR device **600** and VR device **610**.

Example Head-Wearable Devices

[0143] FIGS. 6A, 6B-1, 6B-2, and 6C show example head-wearable devices, in accordance with some embodiments. Head-wearable devices can include, but are not limited to, AR devices **610** (e.g., AR or smart eyewear devices, such as smart glasses, smart monocles, smart contacts, etc.),

VR devices **610** (e.g., VR headsets, head-mounted displays (HMD) s, etc.), or other ocularly coupled devices. The AR devices **600** and the VR devices **610** are instances of the head-wearable devices described in reference to FIGS. **1A-3** herein, such that the head-wearable device should be understood to have the features of the AR devices **600** and/or the VR devices **610**, and vice versa. The AR devices **600** and the VR devices **610** can perform various functions and/or operations associated with navigating through user interfaces and selectively opening applications.

[0144] In some embodiments, an AR system (e.g., AR systems **400a-400d**; FIGS. **4A-4D-2**) includes an AR device **600** (as shown in FIG. **6A**) and/or VR device **610** (as shown in FIGS. **6B-1-B-2**). In some embodiments, the AR device **600** and the VR device **610** can include one or more analogous components (e.g., components for presenting interactive artificial-reality environments, such as processors, memory, and/or presentation devices, including one or more displays and/or one or more waveguides), some of which are described in more detail with respect to FIG. **6C**. The head-wearable devices can use display projectors (e.g., display projector assemblies **607A** and **607B**) and/or waveguides for projecting representations of data to a user. Some embodiments of head-wearable devices do not include displays.

[0145] FIG. **6A** shows an example visual depiction of the AR device **600** (e.g., which may also be described herein as augmented-reality glasses and/or smart glasses). The AR device **600** can work in conjunction with additional electronic components that are not shown in FIGS. **6A**, such as a wearable accessory device and/or an intermediary processing device, in electronic communication or otherwise configured to be used in conjunction with the AR device **600**. In some embodiments, the wearable accessory device and/or the intermediary processing device may be configured to couple with the AR device **600** via a coupling mechanism in electronic communication with a coupling sensor **624**, where the coupling sensor **624** can detect when an electronic device becomes physically or electronically coupled with the AR device **600**. In some embodiments, the AR device **600** can be configured to couple to a housing (e.g., a portion of frame **604** or temple arms **605**), which may include one or more additional coupling mechanisms configured to couple with additional accessory devices. The components shown in FIG. **6A** can be implemented in hardware, software, firmware, or a combination thereof, including one or more signal-processing components and/or application-specific integrated circuits (ASICs).

[0146] The AR device **600** includes mechanical glasses components, including a frame **604** configured to hold one or more lenses (e.g., one or both lenses **606-1** and **606-2**). One of ordinary skill in the art will appreciate that the AR device **600** can include additional mechanical components, such as hinges configured to allow portions of the frame **604** of the AR device **600** to be folded and unfolded, a bridge configured to span the gap between the lenses **606-1** and **606-2** and rest on the user's nose, nose pads configured to rest on the bridge of the nose and provide support for the AR device **600**, earpieces configured to rest on the user's ears and provide additional support for the AR device **600**, temple arms **605** configured to extend from the hinges to the earpieces of the AR device **600**, and the like. One of ordinary skill in the art will further appreciate that some examples of the AR device **600** can include none of the mechanical components described herein. For example, smart contact lenses configured to present artificial-reality to users may not include any components of the AR device **600**.

[0147] The lenses **606-1** and **606-2** can be individual displays or display devices (e.g., a waveguide for projected representations). The lenses **606-1** and **606-2** may act together or independently to present an image or series of images to a user. In some embodiments, the lenses **606-1** and **606-2** can operate in conjunction with one or more display projector assemblies **607A** and **607B** to present image data to a user. While the AR device **600** includes two displays, embodiments of this disclosure may be implemented in AR devices with a single near-eye display (NED) or more than two NEDs.

[0148] The AR device **600** includes electronic components, many of which will be described in more detail below with respect to FIG. **6C**. Some example electronic components are illustrated in

FIG. 6A, including sensors **623-1**, **623-2**, **623-3**, **623-4**, **623-5**, and **623-6**, which can be distributed along a substantial portion of the frame **604** of the AR device **600**. The different types of sensors are described below in reference to FIG. 6C. The AR device **600** also includes a left camera **639A** and a right camera **639B**, which are located on different sides of the frame **604**. And the eyewear device includes one or more processors **648A** and **648B** (e.g., an integral microprocessor, such as an ASIC) that is embedded into a portion of the frame **604**.

[0149] FIGS. **6B-1** and **6B-2** show an example visual depiction of the VR device **610** (e.g., a head-mounted display (HMD) **612**, also referred to herein as an artificial-reality headset, a head-wearable device, a VR headset, etc.). The HMD **612** includes a front body **614** and a frame **616** (e.g., a strap or band) shaped to fit around a user's head. In some embodiments, the front body **614** and/or the frame **616** includes one or more electronic elements for facilitating presentation of and/or interactions with an AR and/or VR system (e.g., displays, processors (e.g., processor **648A-1**), IMUs, tracking emitter or detectors, sensors, etc.). In some embodiments, the HMD **612** includes output audio transducers (e.g., an audio transducer **618-1**), as shown in FIG. **6B-2**. In some embodiments, one or more components, such as the output audio transducer(s) **618-1** and the frame **616**, can be configured to attach and detach (e.g., are detachably attachable) to the HMD **612** (e.g., a portion or all of the frame **616**, and/or the output audio transducer **618-1**), as shown in FIG. **6B-2**. In some embodiments, coupling a detachable component to the HMD **612** causes the detachable component to come into electronic communication with the HMD **612**. The VR device **610** includes electronic components, many of which will be described in more detail below with respect to FIG. 6C.

[0150] FIG. **6B-1** to **6B-2** also show that the VR device **610** one or more cameras, such as the left camera **639A** and the right camera **639B**, which can be analogous to the left and right cameras on the frame **604** of the AR device **600**. In some embodiments, the VR device **610** includes one or more additional cameras (e.g., cameras **639C** and **639D**), which can be configured to augment image data obtained by the cameras **639A** and **639B** by providing more information. For example, the camera **639C** can be used to supply color information that is not discerned by cameras **639A** and **639B**. In some embodiments, one or more of the cameras **639A** to **639D** can include an optional IR cut filter configured to remove IR light from being received at the respective camera sensors.

[0151] The VR device **610** can include a housing **690** storing one or more components of the VR device **610** and/or additional components of the VR device **610**. The housing **690** can be a modular electronic device configured to couple with the VR device **610** (or an AR device **600**) and supplement and/or extend the capabilities of the VR device **610** (or an AR device **600**). For example, the housing **690** can include additional sensors, cameras, power sources, processors (e.g., processor **648A-2**), etc. to improve and/or increase the functionality of the VR device **610**. Examples of the different components included in the housing **690** are described below in reference to FIG. 6C.

[0152] Alternatively or in addition, in some embodiments, the head-wearable device, such as the VR device **610** and/or the AR device **600**, includes, or is communicatively coupled to, another external device (e.g., a paired device), such as an HIPD **7** (discussed below in reference to FIGS. **7A-7B**) and/or an optional neckband. The optional neckband can couple to the head-wearable device via one or more connectors (e.g., wired or wireless connectors). The head-wearable device and the neckband can operate independently without any wired or wireless connection between them. In some embodiments, the components of the head-wearable device and the neckband are located on one or more additional peripheral devices paired with the head-wearable device, the neckband, or some combination thereof. Furthermore, the neckband is intended to represent any suitable type or form of paired device. Thus, the following discussion of neckband may also apply to various other paired devices, such as smart watches, smart phones, wrist bands, other wearable devices, hand-held controllers, tablet computers, or laptop computers.

[0153] In some situations, pairing external devices, such as an intermediary processing device (e.g., an HIPD device **700**, an optional neckband, and/or wearable accessory device) with the head-wearable devices (e.g., an AR device **600** and/or VR device **610**) enables the head-wearable devices to achieve a similar form factor of a pair of glasses while still providing sufficient battery and computation power for expanded capabilities. Some, or all, of the battery power, computational resources, and/or additional features of the head-wearable devices can be provided by a paired device or shared between a paired device and the head-wearable devices, thus reducing the weight, heat profile, and form factor of the head-wearable devices overall while allowing the head-wearable devices to retain its desired functionality. For example, the intermediary processing device (e.g., the HIPD **700**) can allow components that would otherwise be included in a head-wearable device to be included in the intermediary processing device (and/or a wearable device or accessory device), thereby shifting a weight load from the user's head and neck to one or more other portions of the user's body. In some embodiments, the intermediary processing device has a larger surface area over which to diffuse and disperse heat to the ambient environment. Thus, the intermediary processing device can allow for greater battery and computation capacity than might otherwise have been possible on the head-wearable devices, standing alone. Because weight carried in the intermediary processing device can be less invasive to a user than weight carried in the head-wearable devices, a user may tolerate wearing a lighter eyewear device and carrying or wearing the paired device for greater lengths of time than the user would tolerate wearing a heavier eyewear device standing alone, thereby enabling an artificial-reality environment to be incorporated more fully into a user's day-to-day activities.

[0154] In some embodiments, the intermediary processing device is communicatively coupled with the head-wearable device and/or to other devices. The other devices may provide certain functions (e.g., tracking, localizing, depth mapping, processing, storage, etc.) to the head-wearable device. In some embodiments, the intermediary processing device includes a controller and a power source. In some embodiments, sensors of the intermediary processing device are configured to sense additional data that can be shared with the head-wearable devices in an electronic format (analog or digital).

[0155] The controller of the intermediary processing device processes information generated by the sensors on the intermediary processing device and/or the head-wearable devices. The intermediary processing device, like an HIPD **700**, can process information generated by one or more sensors of its sensors and/or information provided by other communicatively coupled devices. For example, a head-wearable device can include an IMU, and the intermediary processing device (neckband and/or an HIPD **700**) can compute all inertial and spatial calculations from the IMUs located on the head-wearable device. Additional examples of processing performed by a communicatively coupled device, such as the HIPD **700**, are provided below in reference to FIGS. 7A and 7B.

[0156] Artificial-reality systems may include a variety of types of visual feedback mechanisms. For example, display devices in the AR devices **600** and/or the VR devices **610** may include one or more liquid-crystal displays (LCDs), light emitting diode (LED) displays, organic LED (OLED) displays, and/or any other suitable type of display screen. Artificial-reality systems may include a single display screen for both eyes or may provide a display screen for each eye, which may allow for additional flexibility for varifocal adjustments or for correcting a refractive error associated with the user's vision. Some artificial-reality systems also include optical subsystems having one or more lenses (e.g., conventional concave or convex lenses, Fresnel lenses, or adjustable liquid lenses) through which a user may view a display screen. In addition to or instead of using display screens, some artificial-reality systems include one or more projection systems. For example, display devices in the AR device **600** and/or the VR device **610** may include micro-LED projectors that project light (e.g., using a waveguide) into display devices, such as clear combiner lenses that allow ambient light to pass through. The display devices may refract the projected light toward a user's pupil and may enable a user to simultaneously view both artificial-reality content and the real

world. Artificial-reality systems may also be configured with any other suitable type or form of image projection system. As noted, some AR systems may, instead of blending an artificial reality with actual reality, substantially replace one or more of a user's sensory perceptions of the real world with a virtual experience.

[0157] While the example head-wearable devices are respectively described herein as the AR device **600** and the VR device **610**, either or both of the example head-wearable devices described herein can be configured to present fully-immersive VR scenes presented in substantially all of a user's field of view, additionally or alternatively to, subtler augmented-reality scenes that are presented within a portion, less than all, of the user's field of view.

[0158] In some embodiments, the AR device **600** and/or the VR device **610** can include haptic feedback systems. The haptic feedback systems may provide various types of cutaneous feedback, including vibration, force, traction, shear, texture, and/or temperature. The haptic feedback systems may also provide various types of kinesthetic feedback, such as motion and compliance. The haptic feedback can be implemented using motors, piezoelectric actuators, fluidic systems, and/or a variety of other types of feedback mechanisms. The haptic feedback systems may be implemented independently of other artificial-reality devices, within other artificial-reality devices, and/or in conjunction with other artificial-reality devices (e.g., wrist-wearable devices which may be incorporated into headwear, gloves, body suits, handheld controllers, environmental devices (e.g., chairs or floormats), and/or any other type of device or system, such as a wrist-wearable device **500**, an HIPD **700**, smart textile-based garment **800**, etc.), and/or other devices described herein.

[0159] FIG. **6C** illustrates a computing system **620** and an optional housing **690**, each of which show components that can be included in a head-wearable device (e.g., the AR device **600** and/or the VR device **610**). In some embodiments, more or less components can be included in the optional housing **690** depending on practical restraints of the respective head-wearable device being described. Additionally or alternatively, the optional housing **690** can include additional components to expand and/or augment the functionality of a head-wearable device.

[0160] In some embodiments, the computing system **620** and/or the optional housing **690** can include one or more peripheral interfaces **622A** and **622B**, one or more power systems **642A** and **642B** (including charger input **643**, PMIC **644**, and battery **645**), one or more controllers **646A** and **646B** (including one or more haptic controllers **647**), one or more processors **648A** and **648B** (as defined above, including any of the examples provided), and memory **650A** and **650B**, which can all be in electronic communication with each other. For example, the one or more processors **648A** and/or **648B** can be configured to execute instructions stored in the memory **650A** and/or **650B**, which can cause a controller of the one or more controllers **646A** and/or **646B** to cause operations to be performed at one or more peripheral devices of the peripherals interfaces **622A** and/or **622B**. In some embodiments, each operation described can occur based on electrical power provided by the power system **642A** and/or **642B**.

[0161] In some embodiments, the peripherals interface **622A** can include one or more devices configured to be part of the computing system **620**, many of which have been defined above and/or described with respect to wrist-wearable devices shown in FIGS. **5A** and **5B**. For example, the peripherals interface can include one or more sensors **623A**. Some example sensors include: one or more coupling sensors **624**, one or more acoustic sensors **625**, one or more imaging sensors **626**, one or more EMG sensors **627**, one or more capacitive sensors **628**, and/or one or more IMUs **629**. In some embodiments, the sensors **623A** further include depth sensors **667**, light sensors **668** and/or any other types of sensors defined above or described with respect to any other embodiments discussed herein.

[0162] In some embodiments, the peripherals interface can include one or more additional peripheral devices, including one or more NFC devices **630**, one or more GPS devices **631**, one or more LTE devices **632**, one or more WiFi and/or Bluetooth devices **633**, one or more buttons **634** (e.g., including buttons that are slidable or otherwise adjustable), one or more displays **635A**, one

or more speakers **636A**, one or more microphones **637A**, one or more cameras **638A** (e.g., including the a first camera **639-1** through nth camera **639-n**, which are analogous to the left camera **639A** and/or the right camera **639B**), one or more haptic devices **640**; and/or any other types of peripheral devices defined above or described with respect to any other embodiments discussed herein.

[0163] The head-wearable devices can include a variety of types of visual feedback mechanisms (e.g., presentation devices). For example, display devices in the AR device **600** and/or the VR device **610** can include one or more liquid-crystal displays (LCDs), light emitting diode (LED) displays, organic LED (OLED) displays, micro-LEDs, and/or any other suitable types of display screens. The head-wearable devices can include a single display screen (e.g., configured to be seen by both eyes), and/or can provide separate display screens for each eye, which can allow for additional flexibility for varifocal adjustments and/or for correcting a refractive error associated with the user's vision. Some embodiments of the head-wearable devices also include optical subsystems having one or more lenses (e.g., conventional concave or convex lenses, Fresnel lenses, or adjustable liquid lenses) through which a user can view a display screen. For example, respective displays **635A** can be coupled to each of the lenses **606-1** and **606-2** of the AR device **600**. The displays **635A** coupled to each of the lenses **606-1** and **606-2** can act together or independently to present an image or series of images to a user. In some embodiments, the AR device **600** and/or the VR device **610** includes a single display **635A** (e.g., a near-eye display) or more than two displays **635A**.

[0164] In some embodiments, a first set of one or more displays **635A** can be used to present an augmented-reality environment, and a second set of one or more display devices **635A** can be used to present a virtual-reality environment. In some embodiments, one or more waveguides are used in conjunction with presenting artificial-reality content to the user of the AR device **600** and/or the VR device **610** (e.g., as a means of delivering light from a display projector assembly and/or one or more displays **635A** to the user's eyes). In some embodiments, one or more waveguides are fully or partially integrated into the AR device **600** and/or the VR device **610**. Additionally, or alternatively to display screens, some artificial-reality systems include one or more projection systems. For example, display devices in the AR device **600** and/or the VR device **610** can include micro-LED projectors that project light (e.g., using a waveguide) into display devices, such as clear combiner lenses that allow ambient light to pass through. The display devices can refract the projected light toward a user's pupil and can enable a user to simultaneously view both artificial-reality content and the real world. The head-wearable devices can also be configured with any other suitable type or form of image projection system. In some embodiments, one or more waveguides are provided additionally or alternatively to the one or more display(s) **635A**.

[0165] In some embodiments of the head-wearable devices, ambient light and/or a real-world live view (e.g., a live feed of the surrounding environment that a user would normally see) can be passed through a display element of a respective head-wearable device presenting aspects of the AR system. In some embodiments, ambient light and/or the real-world live view can be passed through a portion less than all, of an AR environment presented within a user's field of view (e.g., a portion of the AR environment co-located with a physical object in the user's real-world environment that is within a designated boundary (e.g., a guardian boundary) configured to be used by the user while they are interacting with the AR environment). For example, a visual user interface element (e.g., a notification user interface element) can be presented at the head-wearable devices, and an amount of ambient light and/or the real-world live view (e.g., 15-50% of the ambient light and/or the real-world live view) can be passed through the user interface element, such that the user can distinguish at least a portion of the physical environment over which the user interface element is being displayed.

[0166] The head-wearable devices can include one or more external displays **635A** for presenting information to users. For example, an external display **635A** can be used to show a current battery

level, network activity (e.g., connected, disconnected, etc.), current activity (e.g., playing a game, in a call, in a meeting, watching a movie, etc.), and/or other relevant information. In some embodiments, the external displays **635A** can be used to communicate with others. For example, a user of the head-wearable device can cause the external displays **635A** to present a do not disturb notification. The external displays **635A** can also be used by the user to share any information captured by the one or more components of the peripherals interface **622A** and/or generated by head-wearable device (e.g., during operation and/or performance of one or more applications).

[0167] The memory **650A** can include instructions and/or data executable by one or more processors **648A** (and/or processors **648B** of the housing **690**) and/or a memory controller of the one or more controllers **646A** (and/or controller **646B** of the housing **690**). The memory **650A** can include one or more operating systems **651**; one or more applications **652**; one or more communication interface modules **653A**; one or more graphics modules **654A**; one or more AR processing modules **655A**; and/or any other types of modules or components defined above or described with respect to any other embodiments discussed herein.

[0168] The data **660** stored in memory **650A** can be used in conjunction with one or more of the applications and/or programs discussed above. The data **660** can include profile data **661**; sensor data **662**; media content data **663**; AR application data **664**; and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0169] In some embodiments, the controller **646A** of the head-wearable devices processes information generated by the sensors **623A** on the head-wearable devices and/or another component of the head-wearable devices and/or communicatively coupled with the head-wearable devices (e.g., components of the housing **690**, such as components of peripherals interface **622B**). For example, the controller **646A** can process information from the acoustic sensors **625** and/or image sensors **626**. For each detected sound, the controller **646A** can perform a direction of arrival (DOA) estimation to estimate a direction from which the detected sound arrived at a head-wearable device. As one or more of the acoustic sensors **625** detects sounds, the controller **646A** can populate an audio data set with the information (e.g., represented by sensor data **662**).

[0170] In some embodiments, a physical electronic connector can convey information between the head-wearable devices and another electronic device, and/or between one or more processors **648A** of the head-wearable devices and the controller **646A**. The information can be in the form of optical data, electrical data, wireless data, or any other transmittable data form. Moving the processing of information generated by the head-wearable devices to an intermediary processing device can reduce weight and heat in the eyewear device, making it more comfortable and safer for a user. In some embodiments, an optional accessory device (e.g., an electronic neckband or an HIPD **700**) is coupled to the head-wearable devices via one or more connectors. The connectors can be wired or wireless connectors and can include electrical and/or non-electrical (e.g., structural) components. In some embodiments, the head-wearable devices and the accessory device can operate independently without any wired or wireless connection between them.

[0171] The head-wearable devices can include various types of computer vision components and subsystems. For example, the AR device **600** and/or the VR device **610** can include one or more optical sensors such as two-dimensional (2D) or three-dimensional (3D) cameras, time-of-flight depth sensors, single-beam or sweeping laser rangefinders, 3D LiDAR sensors, and/or any other suitable type or form of optical sensor. A head-wearable device can process data from one or more of these sensors to identify a location of a user and/or aspects of the user's real-world physical surroundings, including the locations of real-world objects within the real-world physical surroundings. In some embodiments, the methods described herein are used to map the real world, to provide a user with context about real-world surroundings, and/or to generate interactable virtual objects (which can be replicas or digital twins of real-world objects that can be interacted with in AR environment), among a variety of other functions. For example, FIGS. **6B-1** and **6B-2** show the VR device **610** having cameras **639A-639D**, which can be used to provide depth information for

creating a voxel field and a two-dimensional mesh to provide object information to the user to avoid collisions.

[0172] The optional housing **690** can include analogous components to those describe above with respect to the computing system **620**. For example, the optional housing **690** can include a respective peripherals interface **622B** including more or less components to those described above with respect to the peripherals interface **622A**. As described above, the components of the optional housing **690** can be used augment and/or expand on the functionality of the head-wearable devices. For example, the optional housing **690** can include respective sensors **623B**, speakers **636B**, displays **635B**, microphones **637B**, cameras **638B**, and/or other components to capture and/or present data. Similarly, the optional housing **690** can include one or more processors **648B**, controllers **646B**, and/or memory **650B** (including respective communication interface modules **653B**; one or more graphics modules **654B**; one or more AR processing modules **655B**, etc.) that can be used individually and/or in conjunction with the components of the computing system **620**.

[0173] The techniques described above in FIGS. **6A-6C** can be used with different head-wearable devices. In some embodiments, the head-wearable devices (e.g., the AR device **600** and/or the VR device **610**) can be used in conjunction with one or more wearable device such as a wrist-wearable device **500** (or components thereof) and/or a smart textile-based garment **800** (FIGS. **8A-8C**), as well as an HIPD **700**. Having thus described example the head-wearable devices, attention will now be turned to example handheld intermediary processing devices, such as HIPD **700**.

Example Handheld Intermediary Processing Devices

[0174] FIGS. **7A** and **7B** illustrate an example handheld intermediary processing device (HIPD) **700**, in accordance with some embodiments. The HIPD **700** can perform various functions and/or operations associated with navigating through user interfaces and selectively opening applications.

[0175] FIG. **7A** shows a top view **705** and a side view **725** of the HIPD **700**. The HIPD **700** is configured to communicatively couple with one or more wearable devices (or other electronic devices) associated with a user. For example, the HIPD **700** is configured to communicatively couple with a user's wrist-wearable device **500** (or components thereof, such as the watch body **520** and the wearable band **510**), AR device **600**, and/or VR device **610**. The HIPD **700** can be configured to be held by a user (e.g., as a handheld controller), carried on the user's person (e.g., in their pocket, in their bag, etc.), placed in proximity of the user (e.g., placed on their desk while seated at their desk, on a charging dock, etc.), and/or placed at or within a predetermined distance from a wearable device or other electronic device (e.g., where, in some embodiments, the predetermined distance is the maximum distance (e.g., 10 meters) at which the HIPD **700** can successfully be communicatively coupled with an electronic device, such as a wearable device).

[0176] The HIPD **700** can perform various functions independently and/or in conjunction with one or more wearable devices (e.g., wrist-wearable device **500**, AR device **600**, VR device **610**, etc.). The HIPD **700** is configured to increase and/or improve the functionality of communicatively coupled devices, such as the wearable devices. The HIPD **700** is configured to perform one or more functions or operations associated with interacting with user interfaces and applications of communicatively coupled devices, interacting with an AR environment, interacting with VR environment, and/or operating as a human-machine interface controller. Additionally, as will be described in more detail below, functionality and/or operations of the HIPD **700** can include, without limitation, task offloading and/or handoffs; thermals offloading and/or handoffs; 6 degrees of freedom (6DoF) raycasting and/or gaming (e.g., using imaging devices or cameras **714A** and **714B**, which can be used for simultaneous localization and mapping (SLAM) and/or with other image processing techniques); portable charging; messaging; image capturing via one or more imaging devices or cameras (e.g., cameras **722A** and **722B**); sensing user input (e.g., sensing a touch on a multi-touch input surface **702**); wireless communications and/or interlining (e.g., cellular, near field, Wi-Fi, personal area network, etc.); location determination; financial transactions; providing haptic feedback; alarms; notifications; biometric authentication; health

monitoring; sleep monitoring; etc. The above-example functions can be executed independently in the HIPD **700** and/or in communication between the HIPD **700** and another wearable device described herein. In some embodiments, functions can be executed on the HIPD **700** in conjunction with an AR environment. As the skilled artisan will appreciate upon reading the descriptions provided herein, the novel the HIPD **700** described herein can be used with any type of suitable AR environment.

[0177] While the HIPD **700** is communicatively coupled with a wearable device and/or other electronic device, the HIPD **700** is configured to perform one or more operations initiated at the wearable device and/or the other electronic device. In particular, one or more operations of the wearable device and/or the other electronic device can be offloaded to the HIPD **700** to be performed. The HIPD **700** performs the one or more operations of the wearable device and/or the other electronic device and provides to data corresponded to the completed operations to the wearable device and/or the other electronic device. For example, a user can initiate a video stream using AR device **600** and back-end tasks associated with performing the video stream (e.g., video rendering) can be offloaded to the HIPD **700**, which the HIPD **700** performs and provides corresponding data to the AR device **600** to perform remaining front-end tasks associated with the video stream (e.g., presenting the rendered video data via a display of the AR device **600**). In this way, the HIPD **700**, which has more computational resources and greater thermal headroom than a wearable device, can perform computationally intensive tasks for the wearable device improving performance of an operation performed by the wearable device.

[0178] The HIPD **700** includes a multi-touch input surface **702** on a first side (e.g., a front surface) that is configured to detect one or more user inputs. In particular, the multi-touch input surface **702** can detect single tap inputs, multi-tap inputs, swipe gestures and/or inputs, force-based and/or pressure-based touch inputs, held taps, and the like. The multi-touch input surface **702** is configured to detect capacitive touch inputs and/or force (and/or pressure) touch inputs. The multi-touch input surface **702** includes a first touch-input surface **704** defined by a surface depression, and a second touch-input surface **706** defined by a substantially planar portion. The first touch-input surface **704** can be disposed adjacent to the second touch-input surface **706**. In some embodiments, the first touch-input surface **704** and the second touch-input surface **706** can be different dimensions, shapes, and/or cover different portions of the multi-touch input surface **702**. For example, the first touch-input surface **704** can be substantially circular and the second touch-input surface **706** is substantially rectangular. In some embodiments, the surface depression of the multi-touch input surface **702** is configured to guide user handling of the HIPD **700**. In particular, the surface depression is configured such that the user holds the HIPD **700** upright when held in a single hand (e.g., such that the using imaging devices or cameras **714A** and **714B** are pointed toward a ceiling or the sky). Additionally, the surface depression is configured such that the user's thumb rests within the first touch-input surface **704**.

[0179] In some embodiments, the different touch-input surfaces include a plurality of touch-input zones. For example, the second touch-input surface **706** includes at least a first touch-input zone **708** within a second touch-input zone **706** and a third touch-input zone **710** within the first touch-input zone **708**. In some embodiments, one or more of the touch-input zones are optional and/or user defined (e.g., a user can specific a touch-input zone based on their preferences). In some embodiments, each touch-input surface and/or touch-input zone is associated with a predetermined set of commands. For example, a user input detected within the first touch-input zone **708** causes the HIPD **700** to perform a first command and a user input detected within the second touch-input zone **706** causes the HIPD **700** to perform a second command, distinct from the first. In some embodiments, different touch-input surfaces and/or touch-input zones are configured to detect one or more types of user inputs. The different touch-input surfaces and/or touch-input zones can be configured to detect the same or distinct types of user inputs. For example, the first touch-input zone **708** can be configured to detect force touch inputs (e.g., a magnitude at which the user presses

down) and capacitive touch inputs, and the second touch-input zone **706** can be configured to detect capacitive touch inputs.

[0180] The HIPD **700** includes one or more sensors **751** for sensing data used in the performance of one or more operations and/or functions. For example, the HIPD **700** can include an IMU that is used in conjunction with cameras **714** for 3-dimensional object manipulation (e.g., enlarging, moving, destroying, etc. an object) in an AR or VR environment. Non-limiting examples of the sensors **751** included in the HIPD **700** include a light sensor, a magnetometer, a depth sensor, a pressure sensor, and a force sensor. Additional examples of the sensors **751** are provided below in reference to FIG. 7B.

[0181] The HIPD **700** can include one or more light indicators **712** to provide one or more notifications to the user. In some embodiments, the light indicators are LEDs or other types of illumination devices. The light indicators **712** can operate as a privacy light to notify the user and/or others near the user that an imaging device and/or microphone are active. In some embodiments, a light indicator is positioned adjacent to one or more touch-input surfaces. For example, a light indicator can be positioned around the first touch-input surface **704**. The light indicators can be illuminated in different colors and/or patterns to provide the user with one or more notifications and/or information about the device. For example, a light indicator positioned around the first touch-input surface **704** can flash when the user receives a notification (e.g., a message), change red when the HIPD **700** is out of power, operate as a progress bar (e.g., a light ring that is closed when a task is completed (e.g., 0% to 100%)), operates as a volume indicator, etc.).

[0182] In some embodiments, the HIPD **700** includes one or more additional sensors on another surface. For example, as shown FIG. 7A, HIPD **700** includes a set of one or more sensors (e.g., sensor set **720**) on an edge of the HIPD **700**. The sensor set **720**, when positioned on an edge of the of the HIPD **700**, can be positioned at a predetermined tilt angle (e.g., 26 degrees), which allows the sensor set **720** to be angled toward the user when placed on a desk or other flat surface. Alternatively, in some embodiments, the sensor set **720** is positioned on a surface opposite the multi-touch input surface **702** (e.g., a back surface). The one or more sensors of the sensor set **720** are discussed in detail below.

[0183] The side view **725** of the of the HIPD **700** shows the sensor set **720** and camera **714B**. The sensor set **720** includes one or more cameras **722A** and **722B**, a depth projector **724**, an ambient light sensor **728**, and a depth receiver **730**. In some embodiments, the sensor set **720** includes a light indicator **726**. The light indicator **726** can operate as a privacy indicator to let the user and/or those around them know that a camera and/or microphone is active. The sensor set **720** is configured to capture a user's facial expression such that the user can puppet a custom avatar (e.g., showing emotions, such as smiles, laughter, etc., on the avatar or a digital representation of the user). The sensor set **720** can be configured as a side stereo RGB system, a rear indirect Time-of-Flight (iToF) system, or a rear stereo RGB system. As the skilled artisan will appreciate upon reading the descriptions provided herein, the novel HIPD **700** described herein can use different sensor set **720** configurations and/or sensor set **720** placement.

[0184] In some embodiments, the HIPD **700** includes one or more haptic devices **771** (FIG. 7B; e.g., a vibratory haptic actuator) that are configured to provide haptic feedback (e.g., kinesthetic sensation). The sensors **751**, and/or the haptic devices **771** can be configured to operate in conjunction with multiple applications and/or communicatively coupled devices including, without limitation, a wearable devices, health monitoring applications, social media applications, game applications, and artificial reality applications (e.g., the applications associated with artificial reality).

[0185] The HIPD **700** is configured to operate without a display. However, in optional embodiments, the HIPD **700** can include a display **768** (FIG. 7B). The HIPD **700** can also include one or more optional peripheral buttons **767** (FIG. 7B). For example, the peripheral buttons **767**

can be used to turn on or turn off the HIPD 700. Further, the HIPD 700 housing can be formed of polymers and/or elastomer elastomers. The HIPD 700 can be configured to have a non-slip surface to allow the HIPD 700 to be placed on a surface without requiring a user to watch over the HIPD 700. In other words, the HIPD 700 is designed such that it would not easily slide off a surfaces. In some embodiments, the HIPD 700 include one or magnets to couple the HIPD 700 to another surface. This allows the user to mount the HIPD 700 to different surfaces and provide the user with greater flexibility in use of the HIPD 700.

[0186] As described above, the HIPD 700 can distribute and/or provide instructions for performing the one or more tasks at the HIPD 700 and/or a communicatively coupled device. For example, the HIPD 700 can identify one or more back-end tasks to be performed by the HIPD 700 and one or more front-end tasks to be performed by a communicatively coupled device. While the HIPD 700 is configured to offload and/or handoff tasks of a communicatively coupled device, the HIPD 700 can perform both back-end and front-end tasks (e.g., via one or more processors, such as CPU 777; FIG. 7B). The HIPD 700 can, without limitation, can be used to perform augmenting calling (e.g., receiving and/or sending 3D or 2.5D live volumetric calls, live digital human representation calls, and/or avatar calls), discreet messaging, 6DoF portrait/landscape gaming, AR/VR object manipulation, AR/VR content display (e.g., presenting content via a virtual display), and/or other AR/VR interactions. The HIPD 700 can perform the above operations alone or in conjunction with a wearable device (or other communicatively coupled electronic device).

[0187] FIG. 7B shows block diagrams of a computing system 740 of the HIPD 700, in accordance with some embodiments. The HIPD 700, described in detail above, can include one or more components shown in HIPD computing system 740. The HIPD 700 will be understood to include the components shown and described below for the HIPD computing system 740. In some embodiments, all, or a substantial portion of the components of the HIPD computing system 740 are included in a single integrated circuit. Alternatively, in some embodiments, components of the HIPD computing system 740 are included in a plurality of integrated circuits that are communicatively coupled.

[0188] The HIPD computing system 740 can include a processor (e.g., a CPU 777, a GPU, and/or a CPU with integrated graphics), a controller 775, a peripherals interface 750 that includes one or more sensors 751 and other peripheral devices, a power source (e.g., a power system 795), and memory (e.g., a memory 778) that includes an operating system (e.g., an operating system 779), data (e.g., data 788), one or more applications (e.g., applications 780), and one or more modules (e.g., a communications interface module 781, a graphics module 782, a task and processing management module 783, an interoperability module 784, an AR processing module 785, a data management module 786, etc.). The HIPD computing system 740 further includes a power system 795 that includes a charger input and output 796, a PMIC 797, and a battery 798, all of which are defined above.

[0189] In some embodiments, the peripherals interface 750 can include one or more sensors 751. The sensors 751 can include analogous sensors to those described above in reference to FIG. 5B. For example, the sensors 751 can include imaging sensors 754, (optional) EMG sensors 756, IMUs 758, and capacitive sensors 760. In some embodiments, the sensors 751 can include one or more pressure sensor 752 for sensing pressure data, an altimeter 753 for sensing an altitude of the HIPD 700, a magnetometer 755 for sensing a magnetic field, a depth sensor 757 (or a time-of flight sensor) for determining a difference between the camera and the subject of an image, a position sensor 759 (e.g., a flexible position sensor) for sensing a relative displacement or position change of a portion of the HIPD 700, a force sensor 761 for sensing a force applied to a portion of the HIPD 700, and a light sensor 762 (e.g., an ambient light sensor) for detecting an amount of lighting. The sensors 751 can include one or more sensors not shown in FIG. 7B.

[0190] Analogous to the peripherals described above in reference to FIGS. 5B, the peripherals interface 750 can also include an NFC component 763, a GPS component 764, an LTE component

765, a Wi-Fi and/or Bluetooth communication component **766**, a speaker **769**, a haptic device **771**, and a microphone **773**. As described above in reference to FIG. 7A, the HIPD **700** can optionally include a display **768** and/or one or more buttons **767**. The peripherals interface **750** can further include one or more cameras **770**, touch surfaces **772**, and/or one or more light emitters **774**. The multi-touch input surface **702** described above in reference to FIG. 7A is an example of touch surface **772**. The light emitters **774** can be one or more LEDs, lasers, etc. and can be used to project or present information to a user. For example, the light emitters **774** can include light indicators **712** and **726** described above in reference to FIG. 7A. The cameras **770** (e.g., cameras **714A**, **714B**, and **722** described above in FIG. 7A) can include one or more wide angle cameras, fish-eye cameras, spherical cameras, compound eye cameras (e.g., stereo and multi cameras), depth cameras, RGB cameras, ToF cameras, RGB-D cameras (depth and ToF cameras), and/or other available cameras. Cameras **770** can be used for SLAM; 6 DoF ray casting, gaming, object manipulation, and/or other rendering; facial recognition and facial expression recognition, etc. [0191] Similar to the watch body computing system **560** and the watch band computing system **530** described above in reference to FIG. 5B, the HIPD computing system **740** can include one or more haptic controllers **776** and associated componentry (e.g., haptic devices **771**) for providing haptic events at the HIPD **700**.

[0192] Memory **778** can include high-speed random-access memory and/or non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Access to the memory **778** by other components of the HIPD **700**, such as the one or more processors and the peripherals interface **750**, can be controlled by a memory controller of the controllers **775**.

[0193] In some embodiments, software components stored in the memory **778** include one or more operating systems **779**, one or more applications **780**, one or more communication interface modules **781**, one or more graphics modules **782**, one or more data management modules **785**, which are analogous to the software components described above in reference to FIG. 5B.

[0194] In some embodiments, software components stored in the memory **778** include a task and processing management module **783** for identifying one or more front-end and back-end tasks associated with an operation performed by the user, performing one or more front-end and/or back-end tasks, and/or providing instructions to one or more communicatively coupled devices that cause performance of the one or more front-end and/or back-end tasks. In some embodiments, the task and processing management module **783** uses data **788** (e.g., device data **790**) to distribute the one or more front-end and/or back-end tasks based on communicatively coupled devices' computing resources, available power, thermal headroom, ongoing operations, and/or other factors. For example, the task and processing management module **783** can cause the performance of one or more back-end tasks (of an operation performed at communicatively coupled AR device **600**) at the HIPD **700** in accordance with a determination that the operation is utilizing a predetermined amount (e.g., at least 70%) of computing resources available at the AR device **600**.

[0195] In some embodiments, software components stored in the memory **778** include an interoperability module **784** for exchanging and utilizing information received and/or provided to distinct communicatively coupled devices. The interoperability module **784** allows for different systems, devices, and/or applications to connect and communicate in a coordinated way without user input. In some embodiments, software components stored in the memory **778** include an AR module **785** that is configured to process signals based at least on sensor data for use in an AR and/or VR environment. For example, the AR processing module **785** can be used for 3D object manipulation, gesture recognition, facial and facial expression, recognition, etc.

[0196] The memory **778** can also include data **787**, including structured data. In some embodiments, the data **787** can include profile data **789**, device data **789** (including device data of one or more devices communicatively coupled with the HIPD **700**, such as device type, hardware, software, configurations, etc.), sensor data **791**, media content data **792**, and application data **793**.

[0197] It should be appreciated that the HIPD computing system **740** is an example of a computing system within the HIPD **700**, and that the HIPD **700** can have more or fewer components than shown in the HIPD computing system **740**, combine two or more components, and/or have a different configuration and/or arrangement of the components. The various components shown in HIPD computing system **740** are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application-specific integrated circuits.

[0198] The techniques described above in FIG. 7A-7B can be used with any device used as a human-machine interface controller. In some embodiments, an HIPD **700** can be used in conjunction with one or more wearable device such as a head-wearable device (e.g., AR device **600** and VR device **610**) and/or a wrist-wearable device **500** (or components thereof). In some embodiments, an HIPD **700** can also be used in conjunction with a wearable garment, such as smart textile-based garment **800** (FIGS. 8A-8C). Having thus described example HIPD **700**, attention will now be turned to example feedback devices, such as smart textile-based garment **800**.

Example Smart Textile-Based Garments

[0199] FIGS. 8A and 8B illustrate an example smart textile-based garment, in accordance with some embodiments. The smart textile-based garment **800** (e.g., wearable gloves, a shirt, a headband, a wristbands, socks, etc.) is configured to communicatively couple with one or more electronic devices, such as a wrist-wearable device **500**, a head-wearable device, an HIPD **700**, a laptop, tablet, and/or other computing devices. The smart textile-based garment **800** can perform various functions and/or operations associated with navigating through user interfaces and selectively opening applications.

[0200] The smart textile-based garment **800** can be part of an AR system, such as AR system **400d** described above in reference to FIGS. 4D-1 and 4D-2. The smart textile-based garment **800** is also configured to provide feedback (e.g., tactile or other haptic feedback) to a user based on the user's interactions with a computing system (e.g., navigation of a user interface, operation of an application (e.g., game vibrations, media responsive haptics), device notifications, etc.)), and/or the user's interactions within an AR environment. In some embodiments, the smart textile-based garment **800** receives instructions from a communicatively coupled device (e.g., the wrist-wearable device **500**, a head-wearable device, and HIPD **700**, etc.) for causing the performance of a feedback response. Alternatively, or in addition, in some embodiments, the smart textile-based garment **800** determines one or more feedback responses to provide a user. The smart textile-based garment **800** can determine the one or more feedback responses based on sensor data captured by one or more of its sensors (e.g., sensors **851**; FIG. 8C) or communicatively coupled sensors (e.g., sensors of a wrist-wearable device **500**, a head-wearable device, an HIPD **700**, and/or other computing device).

[0201] Non-limiting examples of the feedback determined by the smart textile-based garment **800** and/or a communicatively coupled device include visual feedback, audio feedback, haptic (e.g., tactile, kinesthetic, etc.) feedback, thermal or temperature feedback, and/or other sensory perceptible feedback. The smart textile-based garment **800** can include respective feedback devices (e.g., a haptic device or assembly **862** or other feedback devices or assemblies) to provide the feedback responses to the user. Similarly, the smart textile-based garment **800** can communicatively couple with another device (and/or the other device's feedback devices) to coordinate the feedback provided to the user. For example, a VR device **610** can present an AR environment to a user and as the user interacts with objects within the AR environment, such as a virtual cup, the smart textile-based garment **800** provides respective response to the user. In particular, the smart textile-based garment **800** can provide haptic feedback to prevent (or, at a minimum, hinder/resist movement of) one or more of the user's fingers from bending past a certain point to simulate the sensation of touching a solid cup and/or thermal feedback to simulate the sensation of a cold or warm beverage.

[0202] Additionally or alternatively, in some embodiments, the smart textile-based garment **800** is configured to operate as a controller configured to perform one or more functions or operations associated with interacting with user interfaces and applications of communicatively coupled

devices, interacting with an AR environment, interacting with VR environment, and/or operating as a human-machine interface controller.

[0203] FIG. 8A shows one or more haptic assemblies **862** (e.g., first through fourth haptic assemblies **862-1** through **862-4**) on a portion of the smart textile-based garment **800** adjacent to a palmar side of the user's hand and FIG. 8B shows additional haptic assemblies (e.g., a fifth haptic assembly **862-5**) on a portion of the smart textile-based garment **800** adjacent to a dorsal side of the user's hand. In some embodiments, the haptic assemblies **862** include a mechanism that, at a minimum, provide resistance when a respective haptic assembly **862** is transitioned from a first state (e.g., a first pressurized state (e.g., at atmospheric pressure or deflated)) to a second state (e.g., a second pressurized state (e.g., inflated to a threshold pressure)). In other words, the haptic assemblies **862** described can transition between a first pressurized state and a second pressurized state to provide haptic feedback to the user. Structures of haptic assemblies **862** can be integrated into various devices configured to be in contact or proximity to a user's skin, including, but not limited to devices such as glove worn devices, body worn clothing device, headset devices. Each of the haptic assemblies **862** can be included in or physically coupled to a garment component **804** of the smart textile-based garment **800**. For example, each of the haptic assemblies **862-1**, **862-2**, **862-3**, . . . **862-N** are physically coupled to the garment **804** are configured to contact respective phalanges of a user's thumb and fingers.

[0204] Due to the ever-changing nature of artificial-reality, the haptic assemblies **862** may be required to transition between the multiple states hundreds, or perhaps thousands of times, during a single use. Thus, the haptic assemblies **862** described herein are durable and designed to quickly transition from state to state. To provide some context, in a first pressurized state, the haptic assemblies **862** do not impede free movement of a portion of the wearer's body. For example, one or more haptic assemblies **862** incorporated into a glove are made from flexible materials that do not impede free movement of the wearer's hand and fingers (e.g., an electrostatic-zipping actuator). The haptic assemblies **862** are configured to conform to a shape of the portion of the wearer's body when in the first pressurized state. However, once in a second pressurized state, the haptic assemblies **862** can be configured to restrict and/or impede free movement of the portion of the wearer's body (e.g., appendages of the user's hand). For example, the respective haptic assembly **862** (or multiple respective haptic assemblies) can restrict movement of a wearer's finger (e.g., prevent the finger from curling or extending) when the haptic assembly **862** is in the second pressurized state. Moreover, once in the second pressurized state, the haptic assemblies **862** may take different shapes, with some haptic assemblies **862** configured to take a planar, rigid shape (e.g., flat and rigid), while some other haptic assemblies **862** are configured to curve or bend, at least partially.

[0205] The smart textile-based garment **800** can be one of a plurality of devices in an AR system (e.g., AR systems of FIGS. 4A-4D-2). For example, a user can wear a pair of gloves (e.g., a first type of smart textile-based garment **800**), wear a haptics component of a wrist-wearable device **500** (FIGS. 5A-5B), wear a headband (e.g., a second type of smart textile-based garment **800**), hold an HIPD **700**, etc. As explained above, the haptic assemblies **862** are configured to provide haptic simulations to a wearer of the smart textile-based garments **800**. The garment **804** of each smart textile-based garment **800** can be one of various articles of clothing (e.g., gloves, socks, shirts, pants, etc.). Thus, a user may wear multiple smart textile-based garments **800** that are each configured to provide haptic stimulations to respective parts of the body where the smart textile-based garments **800** are being worn. Although the smart textile-based garment **800** are described as an individual device, in some embodiments, the smart textile-based garment **800** can be combined with other wearable devices described herein. For example, the smart textile-based garment **800** can form part of a VR device **610** (e.g., a headband portion).

[0206] FIG. 8C shows block diagrams of a computing system **840** of the haptic assemblies **862**, in accordance with some embodiments. The computing system **840** can include one or more

peripheral interfaces **850**, one or more power systems **895** (including charger input **896**, PMIC **897**, and battery **898**), one or more controllers **875** (including one or more haptic controllers **876**), one or more processors **877** (as defined above, including any of the examples provided), and memory **878**, which can all be in electronic communication with each other. For example, the one or more processors **877** can be configured to execute instructions stored in the memory **878**, which can cause a controller of the one or more controllers **875** to cause operations to be performed at one or more peripheral devices of the peripherals interface **850**. In some embodiments, each operation described can occur based on electrical power provided by the power system **895**.

[0207] In some embodiments, the peripherals interface **850** can include one or more devices configured to be part of the computing system **840**, many of which have been defined above and/or described with respect to wrist-wearable devices shown in FIGS. 5A-7B. For example, the peripherals interface **850** can include one or more sensors **851**, such as one or more pressure sensors **852**, one or more EMG sensors **856**, one or more IMUs **858**, one or more position sensors **859**, one or more capacitive sensors **860**, one or more force sensors **861**; and/or any other types of sensors defined above or described with respect to any other embodiments discussed herein. In some embodiments, the peripherals interface can include one or more additional peripheral devices including one or more WiFi and/or Bluetooth devices **868**; an LTE component **869**; a GPS component **870**; a microphone **871**; one or more haptic assemblies **862**; one or more support structures **863** (which can include one or more bladders **864**; one or more manifolds **865**; one or more pressure-changing devices **867**; one or more displays **872**; one or more buttons **873**; one or more speakers **874**; and/or any other types of peripheral devices defined above or described with respect to any other embodiments discussed herein. In some embodiments, computing system **840** includes more or less components than those shown in FIG. 8C.

[0208] In some embodiments, each haptic assembly **862** includes a support structure **863**, and at least one bladder **864**. The bladder **864** (e.g., a membrane) is a sealed, inflatable pocket made from a durable and puncture resistance material, such as thermoplastic polyurethane (TPU), a flexible polymer, or the like. The bladder **864** contains a medium (e.g., a fluid such as air, inert gas, or even a liquid) that can be added to or removed from the bladder **864** to change a pressure (e.g., fluid pressure) inside the bladder **864**. The support structure **863** is made from a material that is stronger and stiffer than the material of the bladder **864**. A respective support structure **863** coupled to a respective bladder **864** is configured to reinforce the respective bladder **864** as the respective bladder changes shape and size due to changes in pressure (e.g., fluid pressure) inside the bladder. The above example haptic assembly **862** is non-limiting. The haptic assembly **862** can include eccentric rotating mass (ERM), linear resonant actuators (LRA), voice coil motor (VCM), piezo haptic actuator, thermoelectric devices, solenoid actuators, ultrasonic transducers, thermo-resistive heaters, Peltier devices, and/or other devices configured to generate a perceptible response.

[0209] The smart textile-based garment **800** also includes a haptic controller **876** and a pressure-changing device **867**. Alternatively, in some embodiments, the computing system **840** is communicatively coupled with a haptic controller **876** and/or pressure-changing device **867** (e.g., in electronic communication with one or more processors **877** of the computing system **840**). The haptic controller **876** is configured to control operation of the pressure-changing device **867**, and in turn operation of the smart textile-based garments **800**. For example, the haptic controller **876** sends one or more signals to the pressure-changing device **867** to activate the pressure-changing device **867** (e.g., turn it on and off). The one or more signals can specify a desired pressure (e.g., pounds-per-square inch) to be output by the pressure-changing device **867**. Generation of the one or more signals, and in turn the pressure output by the pressure-changing device **867**, can be based on information collected by sensors **851** of the smart textile-based garment **800** and/or other communicatively coupled device. For example, the haptic controller **876** can provide one or more signals, based on collected sensor data, to cause the pressure-changing device **867** to increase the pressure (e.g., fluid pressure) inside a first haptic assembly **862** at a first time, and provide one or

more additional signals, based on additional sensor data, to the pressure-changing device **867** to cause the pressure-changing device **867** to further increase the pressure inside a second haptic assembly **862** at a second time after the first time. Further, the haptic controller **876** can provide one or more signals to cause the pressure-changing device **867** to inflate one or more bladders **864** in a first portion of a smart textile-based garment **800** (e.g., a first finger), while one or more bladders **864** in a second portion of the smart textile-based garment **800** (e.g., a second finger) remain unchanged. Additionally, the haptic controller **876** can provide one or more signals to cause the pressure-changing device **867** to inflate one or more bladders **864** in a first smart textile-based garment **800** to a first pressure and inflate one or more other bladders **864** in the first smart textile-based garment **800** to a second pressure different from the first pressure. Depending on the number of smart textile-based garments **800** serviced by the pressure-changing device **867**, and the number of bladders therein, many different inflation configurations can be achieved through the one or more signals and the examples above are not meant to be limiting.

[0210] The smart textile-based garment **800** may include an optional manifold **865** between the pressure-changing device **867**, the haptic assemblies **862**, and/or other portions of the smart textile-based garment **800**. The manifold **865** may include one or more valves (not shown) that pneumatically couple each of the haptic assemblies **862** with the pressure-changing device **867** via tubing. In some embodiments, the manifold **865** is in communication with the controller **875**, and the controller **875** controls the one or more valves of the manifold **865** (e.g., the controller generates one or more control signals). The manifold **865** is configured to switchably couple the pressure-changing device **867** with one or more haptic assemblies **862** of the smart textile-based garment **800**. In some embodiments, one or more smart textile-based garment **800** or other haptic devices can be coupled in a network of haptic device and the manifold **865** can distribute the fluid between the coupled smart textile-based garments **800**.

[0211] In some embodiments, instead of using the manifold **865** to pneumatically couple the pressure-changing device **867** with the haptic assemblies **862**, the smart textile-based garment **800** may include multiple pressure-changing devices **867**, where each pressure-changing device **867** is pneumatically coupled directly with a single (or multiple) haptic assembly **862**. In some embodiments, the pressure-changing device **867** and the optional manifold **865** can be configured as part of one or more of the smart textile-based garments **800** (not illustrated) while, in other embodiments, the pressure-changing device **867** and the optional manifold **865** can be configured as external to the smart textile-based garments **800**. In some embodiments, a single pressure-changing device **867** can be shared by multiple smart textile-based garment **800** or other haptic devices. In some embodiments, the pressure-changing device **867** is a pneumatic device, hydraulic device, a pneudraulic device, or some other device capable of adding and removing a medium (e.g., fluid, liquid, gas) from the one or more haptic assemblies **862**.

[0212] The memory **878** includes instructions and data, some or all of which may be stored as non-transitory computer-readable storage media within the memory **878**. For example, the memory **878** can include one or more operating systems **879**; one or more communication interface applications **881**; one or more interoperability modules **884**; one or more AR processing applications **885**; one or more data management modules **886**; and/or one or more specification-specific modules **887** for [. . .]; and/or any other types of data defined above or described with respect to FIGS. 5A-7B.

[0213] The memory **878** also includes data **888** which can be used in conjunction with one or more of the applications discussed above. The data **888** can include: device data **890**; sensor data **891**; and/or any other types of data defined above or described with respect to FIGS. 5A-7B.

[0214] The different components of the computing system **840** (and the smart textile-based garment **800**) shown in FIGS. **8A-8C** can be coupled via a wired connection (e.g., via busing). Alternatively, one or more of the devices shown in FIGS. **8A-8C** may be wirelessly connected (e.g., via short-range communication signals).

Example System for Knitting Smart Textile-Based Garments

[0215] Attention is now directed to FIG. 9, which illustrates a multi-dimensional knitting machine configured to produce multi-dimensional knitted garments in an automated fashion (e.g., with the needling for any hand knitting or other user intervention after initiating the knitting process, including allowing for having an electronic component automatically knitted as an integrated component of the multi-dimensional knitted garments), in accordance with some embodiments. The multi-dimensional knitting machine **900** is a garment-producing device that is computer controlled and user programmable to allow for complex knitted structures to be produced (e.g., smart textile-based garments **800** (FIGS. 8A-8C); such as gloves, tubular fabrics, fabrics with embedded electronic devices, complex knit patterns, special stretch characteristics, unique pattern structures, multi-thread structures, etc.). The multi-dimensional knitting machine **900** includes a first-axis needle bed **902**, a second-axis needle bed **908**, and N-axis needle bed (indicating more than three needle beds are possible). Each one of these needle beds (e.g., needles **904**, needles **910**, and needles **918**) is configured to use multiple different types of knit patterns (e.g., jersey knits, rib knits, interlock knits, French-terry knits, fleece knits, etc.) based on a programmed sequence providing to the multi-dimensional knitting machine **900**, and variations of these knits can be employed to form a single continuous garment (e.g., a combination of jersey knits and French terry knit and/or a first variation of a jersey knit and a second variation of a jersey knit). In some embodiments, the variations of these knits in a single continuous garment can be done without producing seams (e.g., a seamless wearable device can be produced). In some embodiments, the knitting machine is further configured to layer fabrics to produce multilayered wearable structures (e.g., to house one or more electronic components). In some embodiments, each layer in a multilayered wearable structure can be made from a different fabric, which in one example is produced using a conductive yarn. For example, a two-layer knitted capacitive sensor can be produced using the multi-dimensional knitting machine **900**, where the first layer and the second layer use different thread (e.g., a coated-conductive thread and an uncoated-conductive thread). A plurality of fabric spools (e.g., fabric spools **904**, fabric spools **912**, and fabric spools **920**) can be included for each one of the needle beds. Multiple types of fabric spools can be used for each needle bed allowing for even more complex woven structures (also referred to as garments) to be produced. In some embodiments, the fabric spools can also include elastic thread allowing for stretchable fabrics and/or fabrics with shape memory to be produced.

[0216] Each of the needle beds discussed above can also include one or more non-fabric insertion components (e.g., non-fabric insertion components **906**, non-fabric insertion components **914**, and non-fabric insertion components **922**) that are configured to be used to allow for insertion of non-fabric structures into the needle beds, such that the non-knitted structure can be knitted into the knitted structure, while the knitted structure (e.g., garment) is being produced. For example, non-fabric structures can include flexible printed circuit boards, rigid circuit boards, conductive wires, structural ribbing, sensors (e.g., neuromuscular signal sensors, light sensors, PPG sensors, etc.), etc. In some embodiments, a stitch pattern can be adjusted by the multi-dimensional knitting machine (e.g., in accordance with a programmed sequence of knit instructions provided to the machine) to accommodate these structures, which, in some embodiments, means that these structures are knitted into the fabric, instead of being sewn on top of a knitted fabric. This allows for garments to be lighter, thinner, and more comfortable to wear (e.g., by having fewer protrusions applying uneven pressure to the wearer's skin). In some embodiments, these multi-dimensional knitting machines can also knit knitted structures along either or both of a vertical axis or a horizontal depending on desired characteristics of the knitted structure. Knitting along a horizontal axis means that the garment would be produced from a left side to a right side (e.g., a glove would be produced starting with the pinky finger, then moving to the ring finger, then middle finger, etc. Sewing on the vertical means that the garment is produced in a top-down fashion (e.g., a glove would be produced starting from the top of the tallest finger and move down to the wrist portion of the glove (e.g., as shown by **928** in FIG. 9)). With respect to the glove examples, a reverse

manufacturing process is also contemplated (e.g., knitting a thumb first when knitting on the horizontal and knitting the wrist portions when knitting on the vertical). In some embodiments, the insertion component can feed the non-knitted structure to the knitting machine or, in some other embodiments, the insertion component is fed through the knitting machine with the non-knitted structure. In the latter, the insertion component is not integrated into the garment and is discarded. In some embodiments, the insertion component is not fed at all, but is an integrated component of the multi-dimensional knitting machine that is activated based on a programming knit sequence to then allow for insertion of a non-knitting component into a knitted structure.

[0217] The multi-dimensional knitting machine **900** also includes knitting logic module **924**, which is a module that is user programmable to allow for a user (which can be a manufacturing entity producing wearable structures on mass scale) to define a knitting sequence to produce a garment using any of the above-described materials, stitch patterns, knitting techniques, etc. As stated above, the knitting logic module **924** allows for a seamless combination of any of the above-described techniques, thereby allowing unique complex knitted structures to be produced in a single knitting sequence (e.g., the user does not need to remove the knitted structure, then reinsert and reorient it to complete knitting the knitted structure). The multi-dimensional knitting machine **900** also includes insertion logic module **926**, which works in tandem with the knitting logic module **924**, to allow for insertion of non-fabric components to be seamlessly inserted into the knitted structure while the knitted structure is knitted together. The insertion logic is in communication with the knitting logic to allow for the knit to be adjusted in accordance with where the non-fabric structure is being inserted. In some embodiments, the user need only show where the non-fabric structure is to be inserted in their mock-up (e.g., at a user interface associated with the multi-dimensional knitting machine, which user interface allows for creating and editing a programmed knit sequence) and the knitting logic module **924** and insertion logic module **926** automatically work together to allow for the knitted structure to be produced.

[0218] FIG. **10** illustrates an alternative deployable mechanism used for securing a pair of smart glasses to the collapsible charging case, in accordance with some embodiments. As compared to the deployable mechanism described in reference to FIGS. **1A-3**, the alternative deployable mechanism **1004** comprises fewer components. As illustrated in FIG. **10**, the T-bar **1000** that is used to retain the pair of smart glasses **1002** in conjunction with a saddle **1006** is shown. In some embodiments, the T-bar **1000** is a continuous piece of material. This simplified design aids in the ease of manufacturing. Further details of this deployable mechanism will be described in the portions relating to the subsequent FIGS. **11-12C**.

[0219] FIG. **11** shows a progression of the deployable mechanism transitioning between a closed state, an open state, and an open state with a pair of smart glasses inserted, in accordance with some embodiments. A first pane **1100** shows that the deployable mechanism **1102** being in its undeployed state, which occurs when the collapsible charging case (as shown in FIGS. **1A** and **1B**) is in its collapsed state and not carrying glasses.

[0220] FIG. **11** also shows, in a second pane **1102**, the deployable mechanism **1102** in its deployed state and not yet holding a pair of smart glasses. As shown, a saddle spring **1104** is configured to urge the deployable mechanism **1102** into its deployed state when the collapsible charging case is not collapsed.

[0221] FIG. **11** further shows, in a third pane **1106**, the deployable mechanism **1103** in its deployed state and holding a pair of smart glasses **1108**. The T-bar **1110** is configured to rotate about the axis point **1112** separately from the saddle **1114** to provide force on the pair of smart glasses **1108** (e.g., along the nose pads) as they slide in and lock into place (e.g., coupling with the pogo pins **1116A** and **1116B** (not pictured) with the charging contacts **1118A** and **1118B** (not pictured) of the pair of smart glasses **1108**).

[0222] In the third pane **1106**, the T-bar **1110** moves away from the saddle **1114** when the pair of smart glasses **1108** act upon the T-bar **1110**. The T-bar **1110** when not acted upon by the pair of

smart glasses **1108** returns to its initial position, which occurs as a result of a spring **1120** located within the saddle **1114**. In other embodiments, which will be described below, the spring **1120** can be replaced with either a magnet, a torsion spring, and/or a leaf spring. In some embodiments, the movement of the T-bar **1110** can be controlled by a combination of one or more of a spring bar, a coiled spring, a leaf spring, and a magnet.

[0223] FIGS. **12A-12C** illustrate three types of deployable mechanisms for controlling the movement of the portion that retains a pair of smart glasses, in accordance with some embodiments. While the components described herein are used with the components described in reference to FIGS. **10** and **11**, however, these components are also applicable to the portion described in reference to FIGS. **1A-1B** and **3**.

[0224] FIG. **12A** shows a first deployable mechanism **1200** for controlling the movement of the portion that retains the pair of smart glasses. Moving from top down, FIG. **12A** shows a glasses retention T-bar **1202** that is configured to hold, in part, the pair of smart glasses, and the movement of the T-bar **1202** is controlled primarily by a magnet **1210**. In some embodiments, the T-bar **1202** is produced using metal injection molded (MIM) and can also include an optional derin overmold around the interface portion **1203** that contacts the nose pads of a pair of smart glasses. A saddle **1204** is coupled to the T-bar **1202** such that the hole pass through **1206** of the T-bar **1202** and the hole pass through **1208** of the saddle **1204** are concentric. In some embodiments, the saddle **1204** is produced from polycarbonate (PC).

[0225] FIG. **12A** shows magnet **1210** coupled to a magnet carrier **1212** that is placed between saddle **1204** and a saddle cover **1214**. In some embodiments, the magnet carrier is constructed of stainless steel or other suitable material that will not deform under the magnets load. In some embodiments, the magnet carrier **1212** and the spring **1216** include respective pass-through holes **1218** and **1220** that are concentric with the pass-through hole **1206** of the T-bar **1202** and the pass-through hole **1208** of the saddle **1204**. In some embodiments, the saddle cover **1214** is produced from the same material as the saddle **1204**, which can be comprised of PC.

[0226] A saddle spring **1216** is also shown in FIG. **12A**, which is configured to urge the saddle **1204**, the saddle cover **1214**, the magnet **1210**, magnet carrier **1212**, and the T-bar **1202** upward about their respective pass-through location. The saddle **1204**, the magnet carrier **1212**, the T-bar **1202** and the saddle spring **1216** are rotatably mounted to a pin **1222** that is coupled to a bracket **1224** that is fixed to a collapsible charging case. In some embodiments, the saddle spring **1216** is one of a leaf spring, torsion spring, or coil spring. In some embodiments, the saddle spring **1216** is comprised of stainless steel. In some embodiments, the pin **1222** is comprised stainless steel and the bracket **1224** is comprised of MIM.

[0227] FIG. **12B** shows a second deployable mechanism **1226** for controlling the movement of the portion that retains the pair of smart glasses. Moving from top down, FIG. **12B** shows a glasses retention T-bar **1228** that is configured to hold, in part, the pair of smart glasses, and the movement of the T-bar **1228** is controlled primarily by both a saddle spring **1242** a spring **1234**. In some embodiments, the T-bar **1228** is produced using MIM (optionally colored in black) and can also include an optional overmold around the interface portion that contacts the nose pads of a pair of smart glasses. In some embodiments, the overmold can comprise POM plastic (optionally colored in black). A saddle **1232** is coupled to the T-bar **1228** such that a hole pass through **1233** of the T-bar **1228** and a hole pass through **1236** of the saddle **1232** are concentric. In some embodiments, the saddle **1232** is produced from polycarbonate (PC).

[0228] FIG. **12B** shows a spring **1234** (e.g., a torsion spring) coupled to the T-bar **1228** and urges the T-bar **1228** downward to remain primarily in contact with the saddle **1232**. The spring **1234** performs the same operation as the magnet described in reference to the first deployable mechanism **1200**. In some embodiments, the spring **1234** is a torsion spring, a coil spring, or leaf spring. In some embodiments, the spring **1234** is mounted to a pin **1237** via a pass-through hole **1239**, which is concentric with a pass-through hole **1238** of the T-bar **1228** and a pass-through hole

1233 of a saddle **1232**. In some embodiments, a saddle cover **1240** is coupled to the saddle **1232**, and a charging flexible printed circuit (FPC) **1241** is placed inside. While a charging FPC **1241** is only shown in FIG. **12B**, the deployable mechanisms of FIG. **12A** also includes charging FPC in similar locations. In some embodiments, the saddle cover **1240** is produced from the same material as the saddle **1232**, which can be comprised of PC.

[0229] A saddle spring **1242** is also shown in FIG. **12B**, which is configured to directly urge the saddle **1232** and the saddle cover **1240** upward about their respective pass-through location. The saddle spring **1242** further indirectly urges the T-bar **1228** upward, but does not directly act upon it. The saddle spring **1242** has a greater spring force than the spring **1234** thereby allowing the saddle to be urged up to its non-collapsed state besides the spring **1234** acting against it. The T-bar **1228** has a greater range of motion than the saddle **1232**, which allows for the spring to

[0230] The T-bar **1228** has a greater range of motion than the saddle **1232** (e.g., the T-bar **1228** has a higher angle of rotation about pin **1237** than the angle of rotation of the saddle **1232**), thus the T-bar **1228** can extend further than the saddle **1232** into an extended region. In this extended region, when the T-bar is not in contact with saddle **1232**, the saddle spring **1242** is no longer indirectly acting on the T-bar. In the extended region, however, the spring **1234** solely acts upon the T-bar to urge it downward. In some embodiments, the T-bar **1228** can extend up to 45 more degrees than the saddle **1232**.

[0231] In some embodiments, the spring **1234** is rotatably mounted to the pin **1237** and can optionally include a bushing **1245** on the pin **1237**. The bushing **1245** is used to allow for springs with different sizes due to spring forces and packaging constraints.

[0232] The saddle **1232**, the T-bar **1228** and the saddle spring **1242** are rotatably mounted to pin **1237** that is coupled to a bracket **1244** that is fixed to a collapsible charging case. In some embodiments, the saddle spring **1242** is one of a leaf spring, torsion spring, or coil spring. In some embodiments, the saddle spring **1242** is comprised of stainless steel. In some embodiments, the pin **1236** is comprised stainless steel and the bracket **1224** is comprised of MIM. In some embodiments, the spring **1234** is also rotatably mounted to pin **1236**.

[0233] FIG. **12C** shows a third deployable mechanism **1246** for controlling the movement of the portion that retains the pair of smart glasses. Moving from top down, FIG. **12C** shows a glasses retention T-bar **1248** that is configured to hold, in part, the pair of smart glasses, and the movement of the T-bar **1248** is controlled primarily by a leaf spring **1250** that is integrated with the T-bar **1248**. In some embodiments, the T-bar **1248** is produced from sheet metal and can also include an optional derin overmold around the interface portion that contacts the nose pads of a pair of smart glasses. A saddle **1252** is coupled to the T-bar **1248** such that the hole pass through **1254** of the T-bar **1248** and the hole pass through **1256** of the saddle **1252** are concentric. In some embodiments, the saddle **1256** is produced from polycarbonate (PC).

[0234] The T-bar **1248** surrounds the saddle **1252**, such that the integrated leaf spring **1250** can use saddle **1252** as leverage to return to its default position, i.e., the T-bar **1248** being configured remain primarily in contact with the saddle **1252** (e.g., a major surface of the T-bar is in contact with a major surface of the saddle) when not acted upon by the pair of smart glasses being inserted.

[0235] FIG. **12C** also shows that a saddle cover **1258** is included that encases a portion of the T-bar when assembled. In some embodiments, the saddle cover **1258** is produced from the same material as the saddle **1252**, which can be comprised of PC.

[0236] A saddle spring **1260** is also shown in FIG. **12C**, which is configured to urge the saddle **1252**, the saddle cover **1258**, and the T-bar **1248** upward about their respective pass-through locations. The saddle **1252**, the T-bar **1248** and the saddle spring **1260** are rotatably mounted to a pin **1262** that is coupled to a bracket **1264** that is fixed to a collapsible charging case. In some embodiments, the saddle spring **1260** is one of a leaf spring, torsion spring, or coil spring. In some embodiments, the saddle spring **1260** is comprised of stainless steel. In some embodiments, the pin **1262** is comprised stainless steel and the bracket **1264** is comprised of MIM.

[0237] FIG. 13 illustrates how the collapsible charging case that uses strategically placed magnets to cause the collapsible charging case to “automatically” open once it is partially opened by a user, in accordance with some embodiments. The first pane 1300 illustrates a force 1302 being present on a wing (side portion) 1304 of the collapsible charging case 1306, which holds the wing 1304 in its folded position. In some embodiments, this force is maintained by top portion 1305 when it is folded over the wing 1304. When no force is acting on the wing 1304, it moves to a perpendicular position via a hinge, which is shown in the second pane 1311. In some embodiments, a top portion 1305 is held in place by another magnet 1307 and corresponding magnet(s) located in different areas to keep the collapsible charging case collapsed and the wing 1304 in its folded position. The magnetic force between the other magnet 1307 and the corresponding magnet(s) is greater than the magnetic force of the magnet(s) 1310 in the wing 1304 and the magnets 1312 in a bottom portion 1308.

[0238] Second pane 1311 illustrates that once the collapsible charging case begins to enter the unfolded state, the magnet(s) 1310 in the wing are drawn to the magnets 1312 in the bottom portion 1308 to urge the collapsible charging case 1306 into the fully non-collapsed position. In the non-collapsed position the wing 1304 is perpendicular to the bottom portion 1308. When the case is to be collapsed the reverse interaction occurs, where a force needs to be applied to the wing 1304 to push it back to being substantially parallel with the bottom portion 1308 and portion 1305 is configured to then be folded over to secure it in place. In some embodiments, another side portion is present in the opposite position of the bottom portion and is configured to act in a similar yet mirrored manner to the wing 1304. In some embodiments, these magnets are placed in edges of the surfaces, but can be placed in any suitable location in order to cause the case to self-assemble. In some embodiments other components can be used to cause self-assembly, including springs, which can optionally work in conjunction with magnets.

[0239] FIG. 14 illustrates cutaway views 1400A and 1400B of a wing 1402 rotating about a bottom portion 1404, in accordance with some embodiments. The transition between cutaway view 1400A and cutaway view 1400B shows a fabric cover 1406 moving with a hinged region 1408 and maintaining coverage of the hinged region 1408 throughout the entire range of motion.

[0240] Any data collection performed by the devices described herein and/or any devices configured to perform or cause the performance of the different embodiments described above in reference to any of the Figures, hereinafter the “devices,” is done with user consent and in a manner that is consistent with all applicable privacy laws. Users are given options to allow the devices to collect data, as well as the option to limit or deny collection of data by the devices. A user is able to opt-in or opt-out of any data collection at any time. Further, users are given the option to request the removal of any collected data.

[0241] It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

[0242] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the claims. As used in the description of the embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0243] As used herein, the term “if” can be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting,” that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that

a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” can be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.

[0244] The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain principles of operation and practical applications, to thereby enable others skilled in the art.

Claims

1. A collapsible charging case for a pair of smart glasses, comprising: a foldable component of the collapsible charging case that is configured to operate in at least two states, including: a folded state that partially defines a first internal volume, and an unfolded state that partially defines a second internal volume that is greater than the first internal volume, and the second internal volume being configured to house a pair of smart glasses; a deployable mechanism configured to contact a nose bridge portion of a pair of smart glasses, and the deployable mechanism is configured to operate in at least two states, including: an undeployed state that occurs when the foldable component of the collapsible charging case is in the folded state, and a deployed state that occurs when the foldable component of the collapsible charging case is in the unfolded state, wherein the deployable mechanism is configured to contact a nose bridge portion of the pair of smart glasses in the deployed state.
2. The collapsible charging case for the pair of smart glasses of claim 1, comprising: another foldable component that is on an opposite side of collapsible charging case relative to the foldable component, wherein the other foldable component mirrors a shape of the foldable component.
3. The collapsible charging case for the pair of smart glasses of claim 1, comprising: another foldable component that is perpendicular to the foldable component, and the other foldable component is configured to cause the foldable component to enter the unfolded state when the other foldable component enters the unfolded state.
4. The collapsible charging case for the pair of smart glasses of claim 3, wherein the deployable mechanism is configured to enter the deployed state in response to the other foldable component entering the unfolded state.
5. The collapsible charging case for the pair of smart glasses of claim 1, wherein the deployable mechanism includes a spring and the spring is in an compressed state when the foldable component is in the folded state and an uncompressed state when in the unfolded state.
6. The collapsible charging case for the pair of smart glasses of claim 1, wherein the deployable mechanism includes a retention bracket, and the retention bracket is configured interface with nose pads of a pair of smart glasses to retain the smart glasses within the collapsible charging case.
7. The collapsible charging case for the pair of smart glasses of claim 6, wherein the deployable mechanism includes a nose bridge interface portion that includes one or more charging contacts, and the nose bridge is configured to contact the pair of smart glasses.
8. The collapsible charging case for the pair of smart glasses of claim 1, wherein the deployable mechanism is partially covered by a cover.
9. The collapsible charging case for the pair of smart glasses of claim 1, wherein the collapsible charging case includes a battery and one or more electrical components configured for charging a pair of smart glasses.
10. The collapsible charging case for the pair of smart glasses of claim 1, wherein the folded state has a cross-sectional shape that is rectangular, and the unfolded state has another cross-sectional

shape that is triangular.

11. The collapsible charging case for the pair of smart glasses of claim 1, wherein the pair of smart glasses are a pair of augmented reality glasses.

12. The collapsible charging case for the pair of smart glasses of claim 1, wherein the collapsible charging case includes one or more magnets to maintain a shape of the folded state and the unfolded state.

13. A system, comprising: at least (i) a collapsible charging case and (ii) a pair of smart glasses, wherein the collapsible charging case includes: a foldable component of the collapsible charging case that is configured to operate in at least two states, including: a folded state that partially defines a first internal volume, and an unfolded state that partially defines a second internal volume that is greater than the first internal volume, and the second internal volume being configured to house a pair of smart glasses; a deployable mechanism configured to contact a nose bridge portion of a pair of smart glasses, and the deployable mechanism is configured to operate in at least two states, including: an undeployed state that occurs when the foldable component of the collapsible charging case is in the folded state, and a deployed state that occurs when the foldable component of the collapsible charging case is in the unfolded state, wherein the deployable mechanism is configured to contact a nose bridge portion of the pair of smart glasses in the deployed state.

14. The system of claim 13, wherein the collapsible charging case includes another foldable component that is on an opposite side of collapsible charging case relative to the foldable component, wherein the other foldable component mirrors a shape of the foldable component.

15. The system of claim 13, wherein the collapsible charging case includes another foldable component that is perpendicular to the foldable component, and the other foldable component is configured to cause the foldable component to enter the unfolded state when the other foldable component enters the unfolded state.

16. The system of claim 13, wherein the deployable mechanism includes a retention bracket, and the retention bracket is configured interface with nose pads of a pair of smart glasses to retain the smart glasses within the collapsible charging case.

17. A method of charging of a pair of smart glasses, comprising: receiving a pair of smart glasses in a collapsible charging case, wherein the collapsible charging case includes: a foldable component of the collapsible charging case that is configured to operate in at least two states, including: a folded state that partially defines a first internal volume, and an unfolded state that partially defines a second internal volume that is greater than the first internal volume, and the second internal volume being configured to house a pair of smart glasses; a deployable mechanism configured to contact a nose bridge portion of the pair of smart glasses, and the deployable mechanism is configured to operate in at least two states, including: an undeployed state that occurs when the foldable component of the collapsible charging case is in the folded state, and a deployed state that occurs when the foldable component of the collapsible charging case is in the unfolded state, wherein the deployable mechanism is configured to contact a nose bridge portion of the pair of smart glasses in the deployed state; and in response to receiving the pair of smart glasses at the deployable mechanism, transmitting a charge from the collapsible charging case to the pair of smart glasses.

18. The method of claim 17, wherein the collapsible includes another foldable component that is on an opposite side of collapsible charging case relative to the foldable component, wherein the other foldable component mirrors a shape of the foldable component.

19. The method of claim 17, wherein the collapsible includes another foldable component that is perpendicular to the foldable component, and the other foldable component is configured to cause the foldable component to enter the unfolded state when the other foldable component enters the unfolded state.

20. The method of claim 17, wherein the deployable mechanism includes a retention bracket, and

the retention bracket is configured interface with nose pads of a pair of smart glasses to retain the smart glasses within the collapsible charging case.
